

Supportive Care in Advanced Chronic Kidney Disease

Sara Davison

CHAPTER OUTLINE

DEFINING KIDNEY SUPPORTIVE CARE, 2016	ADVANCE CARE PLANNING, 2026
KEY COMPONENTS OF KIDNEY SUPPORTIVE CARE, 2016	CONSERVATIVE KIDNEY MANAGEMENT, 2030
CULTURALLY COMPETENT SHARED DECISION MAKING, 2016	PALLIATIVE DIALYSIS, 2033
PROGNOSTICATION, 2018	WITHDRAWAL OF DIALYSIS, 2033
SYMPTOM ASSESSMENT AND MANAGEMENT, 2019	EARLY IDENTIFICATION OF PATIENTS MOST LIKELY TO BENEFIT FROM SUPPORTIVE CARE SERVICES, 2034
CHRONIC PAIN MANAGEMENT, 2022	SUMMARY, 2034
SPECIAL CONSIDERATIONS AT THE END OF LIFE, 2025	

KEY POINTS

- “Kidney supportive care” is replacing the term “kidney palliative care” and encompasses services aimed at improving the quality of life (QOL) for chronic kidney disease (CKD) patients, at any age, throughout the continuum of illness, regardless of life expectancy. It can be provided together with therapies intended to prolong life, such as dialysis, and requires culturally sensitive shared decision making to prioritize the components of medical care most important to the patient. This includes careful attention to physical and emotional symptom management and advance care planning.
- Dialysis treatment may address some symptoms, such as fatigue, anorexia, nausea, and vomiting, especially for more robust individuals with less comorbidity, but it may do little to address symptom burden in more frail patients or those with multimorbidity. When taken as a whole, symptom burden is similar in predialysis G5 CKD patients, those treated with conservative kidney management (i.e., those who have chosen nondialysis care), and those on chronic dialysis.
- The Edmonton Symptom Assessment System: Renal (ESAS-r: Renal) is a simple tool to screen for 12 commonly experienced symptoms in CKD using 0–10 visual analog scales.
- The aim of symptom management is to ameliorate symptoms that are burdensome and adversely impact the patient’s QOL as it is not always necessary or possible to resolve them completely. This usually requires that symptoms be treated to $\leq 3/10$.
- Gabapentin can be used to effectively treat neuropathic pain, restless legs, pruritus, and insomnia if used carefully in low doses and titrated slowly to effect.
- Dialysis is unlikely to benefit patients with significant preexisting functional or cognitive impairment and high levels of comorbidity. These patients are often better cared for with conservative kidney management and careful attention to the principles of kidney supportive care.

Despite advances in predialysis care and dialysis technology, people with advanced chronic kidney disease (CKD) continue to have a shortened life expectancy and poor outcomes, including physical, emotional, and spiritual suffering, and low quality of life (QOL).¹ The majority of patients die in acute care facilities, with aggressive care plans in place² and without accessing palliative care services.^{3–6} Current end-of-life care practices are not consistent with patients' preferences.^{7,8}

Supportive care is central to the provision of patient-centered care for patients with chronic diseases. Many countries are placing increased emphasis on the provision of supportive and end-of-life care by “generalist” and community providers as a component of usual care.⁹ Kidney supportive care, therefore, is increasingly being recognized as a core clinical competency for the care of patients with advanced CKD. Unfortunately, nephrologists are often not trained adequately to address multifactorial suffering and address many of the end-of-life challenges inherent in the care of their patients.¹⁰ Consequently, patients with advanced CKD experience significant unmet care needs.^{7,11,12} Training in supportive care for clinicians treating CKD patients is an urgent priority. This chapter aims to serve as an introduction to kidney supportive care and provide some of the foundations for understanding and delivering the key components of kidney supportive care.

DEFINING KIDNEY SUPPORTIVE CARE

The modern hospice movement began in Britain by Dame Cicely Saunders in the 1960s and was concerned with terminal care, predominantly for cancer patients. Dr. Balfour Mount, a Canadian physician, who opened one of the first two hospice programs in Canada in 1975, coined the term “palliative care” to replace the word “hospice,” which in Canada was commonly considered a place of last resort for the derelict. However, these programs continued to focus predominantly on the care of imminently dying cancer patients. Palliative medicine has undergone many developments since then and has extended the range of services, the timing of services within the illness trajectory, and the eligible patient groups such that palliative care is no longer limited to terminal care focused primarily on those dying with cancer. The WHO defines palliative care as “an approach that improves the QOL of patients and their families facing the problems associated with life-threatening illness, through the prevention and relief of suffering by means of early identification and impeccable assessment and treatment of pain and other problems, physical, psychosocial and spiritual.”¹³

Unfortunately, despite the evolution of palliative care over the past several decades, the belief that “palliative care” and “terminal care” are synonymous, such that only patients at the end of life are appropriate for palliative care, remains prevalent among patients, family, and health-care providers.^{14,15} This is not appropriate for patients with CKD who frequently have high palliative care needs for years before death. For this reason, the term “supportive care” has emerged as yet another term to describe noncurative treatments focused on improving QOL and is increasingly being used to replace the term “palliative care,” until the more terminal phase of life. Both patients and health-care professionals appear to prefer

this.^{16–18} The international kidney community, therefore, is increasingly using the term, “kidney supportive care,” rather than “kidney palliative care.”¹ The term “kidney supportive care” is used throughout this chapter, with the understanding that it conceptually presents the World Health Organization (WHO) definition for palliative care. Kidney supportive care, therefore, is defined as services aimed at improving the QOL for patients with established CKD, at any age, throughout the continuum of illness, regardless of life expectancy and can be provided together with therapies intended to prolong life, such as dialysis.¹ Kidney supportive care is not restricted to withdrawal of dialysis or conservative (nondialysis) kidney management. End-of-life care (sometimes referred to as “terminal care” or “hospice care”) is under the umbrella of supportive care but is typically limited to the care of patients who are believed to be within months of death (Fig. 62.1). Ideally, supportive care should be started early so that issues of suffering are addressed as they present. As illness progresses, the need for supportive care services is likely to increase and may progress into terminal, end-of-life care.

KEY COMPONENTS OF KIDNEY SUPPORTIVE CARE

Kidney supportive care is patient-centered care that integrates culturally sensitive shared decision making to (1) prioritize the components of medical care most important to the patient and (2) ensure those priorities guide clinical decisions.^{19–23} Kidney supportive care includes relief of suffering through meticulous symptom assessment and management; emotional, social and spiritual support; patient-specific estimates of prognosis; advance care planning; and the consideration of treatment options such as conservative kidney management (CKM)—that is, nondialysis care—and the appropriate and timely withdrawal of dialysis. Recent patient engagement has highlighted many of these issues as top priorities for people with advanced CKD.²⁴ A comprehensive approach to kidney supportive care must also integrate a way to identify patients most likely to benefit from supportive care interventions (Fig. 62.2). Each of these components of kidney supportive care is discussed within the context of CKD.

CULTURALLY COMPETENT SHARED DECISION MAKING

Culturally competent shared decision making is the foundation of kidney supportive care and must be incorporated into all components from assessment and management of suffering, sharing prognosis, advance care planning, through to delivery of end-of-life care and bereavement support (see Fig. 62.2). To relieve suffering and provide care aligned to the preferences and goals of the individual patient, care teams must understand and incorporate the patient's needs and perspective and adapt the care plan to facilitate integration of the patient's lifestyle, including his or her family and social community. This means allowing the components of medical care deemed most important to the patient to be prioritized.^{19–23} This may require balancing the management of symptoms (e.g., dizziness and fatigue) with optimal control of blood pressure or serum chemistry with greater emphasis

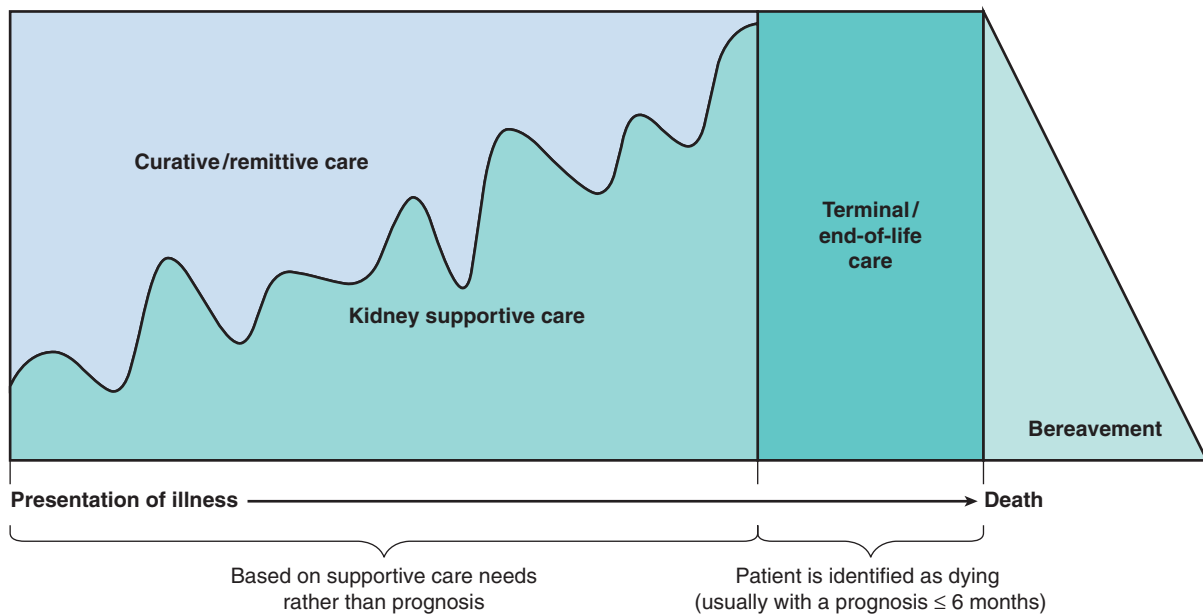


Fig. 62.1 Conceptual framework for kidney supportive care.

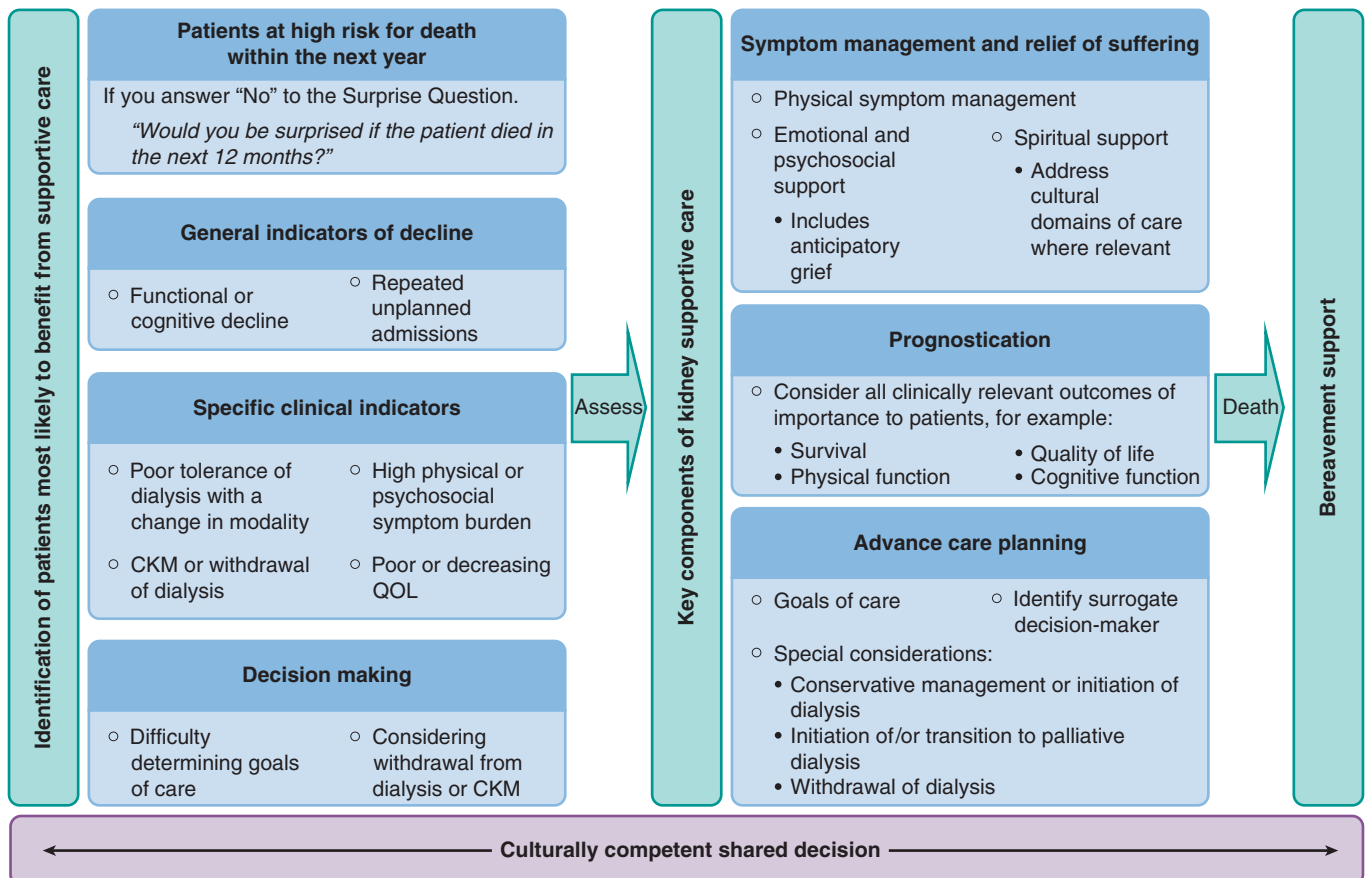


Fig. 62.2 An approach to kidney supportive care. CKM, Conservative kidney management; QOL, quality of life. (Adapted from Davison SN. Integrating palliative care for patients with advanced chronic kidney disease: recent advances, remaining challenges. *J Palliat Care* 2011;27:53–61.)

being placed on managing bothersome symptoms rather than maximizing long-term health outcomes such as survival. As disease progresses, patients' goals of care often shift to focus almost exclusively on QOL rather than survival, with a strong emphasis on emotional, social, and family support.^{7,25} The

ethical imperatives of shared decision are discussed further in Chapter 82.

All cultures generate explanatory models that attempt to account for the phenomenon of illness; to define what a disease is, how it occurs, what measures can prevent or control

it, and why some people and not others are affected. Culture, therefore, has enormous potential to influence patients' beliefs around life and death, healing and suffering, as well as the physician–patient relationship.²⁶ Cultural sensitivity is particularly important in delivering end-of-life care and cultural variations in end-of-life preferences have long been noted.²⁷ In multicultural societies, very different, and often divergent, value systems may be at work and health-care providers will need to be cognizant of these cultural influences. Although a more detailed exploration of how various cultural perspectives may influence the provision of kidney supportive care can be found elsewhere,²⁶ several high-level issues are discussed here.

There are tremendous cross-cultural differences in preferences for decision making. Western cultures tend to promote “patient autonomy,” where the patient is viewed as the best person to make health-care decisions. However, many cultures are characterized by strong communal and social bonds where the person is viewed as a “relational-self,” where social relationships, rather than individualism, provide the basis for moral judgments.²⁶ From this perspective, an insistence on self-determination erodes the value placed on personal interconnectedness and challenges the assumption that the patient should make his or her own medical decisions in isolation from their community. In practice, this might mean that the family or community receive and disclose information, even when the patient is competent.

In many parts of the world, the cultural norm is protection of the patient from the truth. This often involves cultural beliefs surrounding the cause and meaning of illness. This can complicate decision making, especially in the context of sharing prognosis. For example, some Aboriginal and Asian cultures prohibit explicit references to dying based on an interpretative framework in which language has the capacity to create reality.²⁸ Positive thinking is felt to promote health, whereas the delivery of bad news could shorten the life of the patient. Family members may prefer to communicate prognostic information themselves in such a way as to “balance” hope with the bad news. Health-care providers need to understand that in some contexts, this may be appropriate.

Cultural views can impact how patients want care to be delivered. Many cultures possess little knowledge of, or have little exposure to, palliative care and may be reluctant to receive this care without understanding the direct benefit to them. Cultural values may also determine who should and who should not be directly involved in providing physical care, with many cultures strongly preferring that family be directly involved.²⁹ Many non-Western cultures such as Traditional Chinese and indigenous cultures view the body, soul, and spirit as an integrated whole in the context of strong interpersonal relationships and may require access to a traditional healer. This may be viewed as similar to access to a hospital chaplain in a Western context.³⁰

Many people from rural and remote areas wish to die at home connected to land and family for strong cultural reasons, yet are relocated for end-of-life care. For indigenous people, relocation at the end of life is an extremely frightening experience.³¹ Culture also plays a role in the expression of suffering and grief, with many cultures being reluctant to complain of pain. Others are not comfortable with frank, direct styles of communication and therefore may not express suffering or grief overtly as this may be seen as inappropriate.

In these cases, emotional containment does not mean indifference or a lack of suffering.

Culturally competent shared decision making requires that clinicians and patients jointly consider best clinical evidence in light of a patient's specific health characteristics and values within the framework of the patient's and family's cultural beliefs. In increasingly diverse societies, the challenge for health-care professionals is to understand how these numerous cultural differences shape the care for an individual patient. Cross-cultural communication strategies required to address this have recently been explored^{32,33} and are summarized in [Table 62.1](#). Cross-cultural communication starts with understanding one's own beliefs, values, and experiences and then acknowledging individual cultural traditions, which may be sharply divergent. Health-care professionals also need to recognize the diversity of beliefs and practices within cultural groups. The process of acculturating may complicate this and many people will hold blended cultural perspectives. Health-care professionals, therefore, must be careful not to assume the preferences of individuals based on their cultural group. When uncertain about how a patient or family perceives a situation, it is best simply to ask.^{32,33}

PROGNOSTICATION

Estimating and communicating prognosis to patients and family, balancing biomedical facts with relevant emotional, social, cultural, and spiritual issues, are required for shared decision making and providing supportive care.¹ Being able to prognosticate accurately helps health-care professionals identify high-risk patients, facilitate discussions regarding the need for targeted supportive care services, and align care pathways to best meet patients' needs. Given that supportive care focuses on the patient's priorities, patient-specific prognoses for relevant outcomes beyond survival, such as the impact of treatments on QOL, physical function, hospital-free survival, and symptoms, are needed. This would facilitate recommendations such as dialysis services to those most likely to benefit or CKM for those unlikely to benefit. For further discussions on appropriateness of initiating or forgoing dialysis, see Chapters 82 and 84.

Studies have shown that despite patients wanting to discuss their prognosis, it is often not done.^{7,34} Prognostication is challenging in CKD given the great variation in individual illness trajectories.³⁵ Accurate prognostic models are limited, and nephrologists struggle to explain illness complexity and predict clinical trajectories.^{36,37} Although there is a large volume of data on prognostic markers, individually these factors are often of little practical relevance. Only a small number of studies have attempted to combine these factors into clinically useful prognostic tools. A recent review highlights the limits and deficiencies of currently available prognostic tools,³⁸ which are outlined in [Table 62.2](#). These tools are limited to progression of kidney failure and survival on dialysis. They do not prognosticate outcomes with non-dialysis (CKM) care, nor do they address patient or family concerns of QOL, function, and symptom burden.

As the field of prognostication moves forward, new markers may enhance current approaches. These may include identifying a decline in physical function using tools such as gait speed, a modified Karnofsky activity scale, activities of daily

Table 62.1 Strategies for Culturally Competent Shared Decision Making

Task	Communication Strategy	Additional Considerations
Understand one's own beliefs, values, and experiences	Having an awareness of individual differences from that of the patient is important to create a respectful and nonbiased dialog. It will also help resolve conflict when differences with patients' beliefs and wishes arise.	Consider one's perspective as a clinician and the influence of the health-care system and institution within which one delivers care.
Understand the patient's experience	Ask about the patient as a person, where they are from and degree of integration within the ethnic community. Assess how the patient interprets his or her condition (i.e., the cause and impact).	This includes asking what their most important concerns are now that they have this illness. This establishes the groundwork for negotiating mutually acceptable goals for care.
Acknowledge individual cultural traditions	Ask about the patient's health beliefs, values, practices, and cultural communication etiquette. This includes attitudes toward truth telling. Avoid generalizing a patient's beliefs or values based on cultural norms.	The "RISK" reduction assessment is a helpful strategy to ascertain the level of cultural influence for a patient. ³³
Giving information	Ask-Tell-Ask (this strategy can extend to many of the tasks described here): Ask the patient's understanding of his or her illness, what kind of information the patient and/or family needs, and how they want the information told. Tell (give) information concisely, avoid jargon, and disclose only one to three pieces of information at any given time. Ask what the patient understands or will take away from the conversation.	The final ask ensures the information has been understood and invites the patient and family to share concerns or explore lingering questions. Are there language barriers or low health literacy barriers that may hinder understanding and shared decision making?
Determine the level of patient engagement in decision making	Assess preferences for decision making and who will be involved.	Explicitly explore preferences for family-centered versus individual-centered decision making.
Address trust concerns	Be nonjudgmental, transparent, and avoid defensiveness.	The goal is to create an atmosphere of mutual respect and avoid misunderstanding.
Address resources needs	Ask what tangible resources they need to navigate the health-care system. Ask if assistance is available to them and their family in their community.	This may be influenced by language, level of education, socioeconomic status, and their social support networks.

From Brown EA, Bekker HL, Davison SN, et al. Supportive care: communication strategies to improve cultural competence in shared decision making. *Clin J Am Soc Nephrol*. 2016;11:1902–1908.

living, or frailty scores.^{39–45} Repeat hospitalizations,⁴⁶ a decline in QOL scores,⁴⁷ reduced nutritional scores,⁴⁸ reduced appetite,⁴⁹ and lower body weight^{50,51} may all be simple and reliable means for independently identifying CKD patients at risk for early death or poor health outcomes. Ideally, these prognostic tools would be used alongside patient decision aids to help in the shared decision making process. Well-designed decision aids can improve people's knowledge regarding treatment options, reduce their decisional conflict related to feeling uninformed or unclear about their personal values, and can improve accurate risk perceptions.⁵² Accurate prognostication will likely provide better opportunities to meet patients' and families' needs in the final years and months of life. New models, however, will need to be evaluated in different clinical and cultural contexts.

SYMPTOM ASSESSMENT AND MANAGEMENT

Symptom assessment and management are integral components of kidney supportive care. Advanced CKD is associated with a high symptom burden and patients often experience

complex clusters of symptoms, such as pruritus, pain, restless legs, fatigue, anorexia, nausea, insomnia, anxiety, and depression.^{53–55} Although dialysis treatment may address some symptoms, such as fatigue, anorexia, nausea, and vomiting, especially for more robust individuals with less comorbidity, it may do little to address these and other symptoms in more frail patients, and may add to their overall symptom burden. When taken as a whole, symptom burden is similar in pre-dialysis G5 CKD patients and chronic dialysis patients. Research suggests that symptom burden is more important than objective clinical parameters in determining QOL in CKD patients.⁵⁶ Symptom burden was shown to account for 29%–38.5% of the impairment in dialysis patients' physical QOL scores and 39%–42.5% of their impairment in mental QOL scores.⁵³ Moreover, changes in symptom burden were shown to account for 34%–44.6% of the change in physical QOL and 46%–48.7% of the change in mental QOL.^{54,55} Kidney Disease Improving Global Outcomes recommends incorporating regular global symptom screening using validated tools such as the Edmonton Symptom Assessment System: Renal (ESAS-r: Renal)⁵⁷ into routine clinical practice, and the incorporation of systematic, stepwise approaches to managing these symptoms.¹

Table 62.2 Prognostic Tools Relevant to Kidney Supportive Care

Data Source	Population Studied	Parameter	C-Statistic
Progression of CKD to ESKD Defined as Need for Dialysis or Preemptive Kidney Transplantation: Kidney Failure Risk Equation			
Two Canadian cohorts ^{93,94} A total of 31 multinational cohorts from the CKD Prognosis Consortium ⁷¹	Development and initial validation with 3449 and 4942 Canadian patients with G3–5 CKD, respectively. Additional validation in 721,357 multinational participants with G3–5 CKD	Four-variable model: age, gender, eGFR, and ACR Currently available for free online (https://www.qxmd.com/calculate/calculator_308/kidney-failure-risk-equation-4-variable) Eight-variable model: age, gender, eGFR, ACR, calcium, phosphate, bicarbonate, and albumin	Pooled data from the 31 multinational cohorts 2-year prediction: 0.9 (95% CI 0.89–0.92) 5-year prediction: 0.88 (95% CI 0.86–0.90) Pooled data from the 31 multinational cohorts 2-year prediction: 0.89 (95% CI 0.88–0.91) 5-year prediction: 0.87 (95% CI 0.85–0.88)
3 Months' Survival After Dialysis Start			
USRDS ⁹⁵	69,441 incident patients aged ≥ 67 years	Age, albumin, assistance with ADL, nursing home, cancer, heart failure, and hospitalization Currently available for free online (https://qxmd.com/calculate/3-month-mortality-in-incident-elderly-esrd-patients) A more comprehensive version adds gender, race, central vein dialysis catheter use, early nephrology referral, albumin, creatinine, peripheral vascular disease, and alcohol abuse	0.681 (95% CI 0.676–0.686) 0.712 (95% CI 0.706–0.718)
French REIN registry ⁹⁶	28,496 incident patients aged ≥ 75 years	Gender, age, congestive heart failure, severe peripheral vascular disease, dysrhythmia, severe behavioral disorders, active malignancy, serum albumin, and impaired mobility	0.749 (95% CI 0.743–0.755)
Catalan ⁹⁷ renal registry	1365 incident diabetic adult patients	Age, functional autonomy, heart disease, and central catheter as vascular access	0.77 (95% CI 0.742–0.798)
6 Months' Survival After Dialysis Start			
French REIN ⁹⁸ registry	4142 incident patients aged ≥ 75 years	BMI, diabetes, congestive heart failure, peripheral vascular disease, dysrhythmia, active malignancy, severe behavioral disorder, dependency for transfers, and initial context of dialysis start	0.70 (95% CI 0.671–0.729)
6 Months' Survival on Hemodialysis			
New England HD clinics ⁹⁹	1026 adult-prevalent patients on maintenance hemodialysis	Age, dementia, peripheral vascular disease, serum albumin, and the “surprise question” Currently available for free online (https://qxmd.com/calculate/calculator_135/6-month-mortality-on-hd)	0.80 (95% CI 0.73–0.88)

ACR, Urine albumin-to-creatinine ratio; ADL, activities of daily living; BMI, body mass index; CI, confidence interval; CKD, chronic kidney disease; C-statistic, concordance (C)-statistic (which corresponds to the area under receiver operating characteristic curve, ranges from 0 to 1 depending on accuracy of discrimination); eGFR, estimated glomerular filtration rate; ESKD, end-stage kidney disease; G3–5, estimated glomerular filtration rate category 3–5; HD, hemodialysis; REIN, Renal Epidemiology and Information Network; USRDS, United States Renal Data System.

Many global symptom assessment tools of varying length have been used in CKD studies.⁵⁸ For clinical utility it is important that these tools are not only valid and reliable but that they are also sufficiently short and simple to minimize patient and staff burden. They must be appropriate for use in high-risk patients who are often frail with cognitive impairment. In the case of pain, 0–100-mm visual analog scales (VASs) or 0–10 numerical rating scales (NRSs) are advised based on extensive literature around what constitutes clinically important differences in pain intensity.⁵⁹ Core domains that

should be assessed when managing pain include pain intensity, function, QOL, and adverse events.^{59,60}

Several self-reported QOL measures have also been used for CKD patients. Some are generic measures, whereas others are disease specific such as the Kidney Dialysis Quality of Life (KDQOL) Questionnaire and the shorter version (KDQOL-SF).⁶¹ These multidimensional tools focus on physical and emotional symptoms, burden of disease and effects on daily life, cognitive function, work status, sexual function, quality of social interaction and social support,

staff encouragement, and patient satisfaction. Unfortunately, they are burdensome. They typically require interviewer assistance and take substantial time to complete, especially for elderly, frail patients. Although such tools provide comprehensive QOL information, they are perhaps more suited to a research environment where dedicated staff can help with the administration and complex scoring. To embed a QOL measure into routine clinical care, simple tools that can be interpreted easily by staff will most likely be required.^{62,63}

Generic and disease-specific measures have been shown to perform similarly in assessing changes in QOL in hemodialysis patients,⁶⁴ and a recent Oxford review of patient-reported outcome measures in CKD recommended EQ-5D-5L for use in this patient population.⁶⁵ Recommendations for symptom assessment tools that can be used in routine clinical care are outlined in Table 62.3.

The aim of treatment is to ameliorate symptoms that are burdensome and adversely impact the patient's QOL as it is

Table 62.3 Recommendations for Clinical Symptom Assessment Tools for Use in Patients With Advanced CKD

Description	Clinical Utility
Global Symptom Assessment	
Edmonton Symptom Assessment System-Revised: Renal (ESAS-r: Renal)⁵⁷	
A 0–10 NRS for 12 symptoms: pain, activity, nausea, depression, anxiety, drowsiness, appetite, well-being, shortness of breath, pruritus, sleep, and restless legs. It also has a place to capture “other problems.” The scale for each symptom is anchored by the words “no” and “severe” at 0 and 10, respectively. The sum of all scores makes up the overall symptom distress score ranging from 0 to 120.	This is a short, practical tool for global symptom screening, which can be rapidly and repeatedly completed by patients and therefore incorporated easily into routine clinical care, even for patients who are preterminal. It has been translated into several languages. It uses the 0–10 NRS advised for the assessment of pain intensity.
Palliative Care Outcome Scale-Renal (POS-Renal)¹⁰⁰	
Assesses 17 symptoms: pain, shortness of breath, weakness or lack of energy, nausea, vomiting, poor appetite, constipation, mouth problems, drowsiness, poor mobility, itching, difficulty sleeping, restless legs or difficulty keeping legs still, anxiety, depression, changes in skin, and diarrhea. These are rated in terms of their impact on the patient over the last week using a 0 (not at all) to 4 (overwhelmingly) Likert scale.	This tool is slightly longer than the ESAS-r: Renal but is simple to use. It has been translated into several languages. A disadvantage is that it does not use the 0–10 NRS that is advised for assessing pain.
Multidimensional Pain Assessment	
The Brief Pain Inventory (BPI)¹⁰¹	
Assesses the location, type (nociceptive vs. neuropathic), and intensity of pain. It also evaluates the impact of pain on the core domains of function and QOL. Specifically, it explores the impact of pain on general activity, mood, walking ability, work, relationships, sleep, and enjoyment of life. The standard 32-question instrument has been condensed to a nine-question short-form.	This tool has been used successfully in clinical and research settings internationally to assess pain once identified as problem. The short-form is simple to use with minimal respondent burden and seriously ill patients have been successful in completing it. The Interference Scale of the BPI has been recommended by IMMPACT for the assessment of physical functioning, one of the core pain assessment domains. ¹⁰² A change of 1 point on the Interference Scale is considered the minimally clinically important change. ^{59,60} It has been translated into several languages.
Quality of Life Assessment	
5-Level EuroQol 5 Dimension (EQ-5D-5L)⁶⁵	
This generic QOL measure has two parts: (1) EQ-5D descriptive system and (2) EQ VAS. The descriptive system comprises five dimensions: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression. Each dimension has five levels: “no problems,” “slight problems,” “moderate problems,” “severe problems,” and “extreme problems.” The EQ VAS records the patient's self-rated health on a vertical 0–100-mm VAS, where the endpoints are labeled “The best health you can imagine” and “The worst health you can imagine.”	This is a short, practical tool for assessing QOL. It is available in more than 130 languages.

CKD, Chronic kidney disease; IMMPACT, Initiative on Methods, Measurement, and Pain Assessment in Clinical Trials; NRS, numerical rating scale; QOL, quality of life; VAS, visual analog scale.

not always necessary or possible to resolve them completely. It is important to acknowledge this and negotiate with the patient an acceptable level of symptom control. For pain, there is good evidence to suggest that targeting a final pain state of less than 33/100 mm on a VAS or 3/10 on an NRS or a change in pain intensity of at least 30% (moderate benefit) or 50% (substantial benefit) will deliver worthwhile QOL benefits.^{60,66} Given the synergistic and interrelated nature of symptoms experienced in advanced CKD, an approach to care that addresses overall symptom burden is likely to improve QOL even if each individual symptom has not resolved completely. For example, a moderate reduction in pain may be sufficient to improve sleep, improve mood, and increase the ability to cope with health challenges, resulting in a substantial improvement in function and QOL. Recommendations for

the management of common symptoms in CKD are outlined in Table 62.4. These recommendations are appropriate for patients with G5 CKD who have yet to start dialysis, patients who have chosen CKM, and for dialysis patients.^{67,68} A general stepwise approach to symptom management involves (1) ruling out contributing factors, (2) maximizing the use of nonpharmacologic interventions, and finally (3) considering pharmacologic interventions if symptoms continue to impact adversely the patient's QOL.

CHRONIC PAIN MANAGEMENT

Although detailed information on the management of chronic or persistent pain is beyond the scope of this chapter, some

Table 62.4 Symptom Management for Patients With Advanced Chronic Kidney Disease⁶⁷

Restless Legs Syndrome

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
- Anemia - Iron deficiency - Hyperphosphatemia - Medications such as dopamine antagonists, antidepressants, and opioids (some of these drugs are commonly prescribed at end of life, e.g., haloperidol and opioids)	- Abstinence from stimulants (e.g., alcohol, caffeine, and nicotine) - Mental alerting activities (e.g., puzzles or games) - Good sleep hygiene - Exercise	First line: gabapentin (50–300 mg daily) 2–3 hours before sleep, especially if concomitant pruritus, insomnia, and/or neuropathic pain are reported Second line: nonergot-derived dopamine agonists (pramipexole 0.125 mg daily, ropinirole 0.25 mg daily, rotigotine transdermal patch 1–3 mg) given 2 hours before sleep At the end of life if swallowing is problematic: consider midazolam 1 mg subcutaneously q4h PRN	The most common side effects of gabapentin are drowsiness, dizziness, confusion, fatigue, and occasionally peripheral edema Nonergot-derived dopamine agonists have shown success in reducing symptoms in idiopathic RLS, but there are very limited data in uremic RLS. Side effects might include headache, insomnia, and nausea. Augmentation may occur with long-time use. Benzodiazepines are not a first-line treatment for RLS but there is some limited evidence for their use. If the patient is experiencing refractory RLS causing significant sleep disturbance, or if benzodiazepines may potentially treat concurrent symptoms (e.g., anxiety), they could be considered.

Uremic Pruritus

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
- Anemia - Iron deficiency - Hyperphosphatemia - Hypercalcemia - Other: xerosis, drug hypersensitivities, allergies, infestations, contact dermatitis, or inflammation	- Good skin care and moisturizers (e.g., baths with lukewarm water, pat dry and moisturize within 2 minutes; gentle soaps with no fragrances or additives) - Keep skin cool - Humid environment - Avoid scratching—keep fingernails short, encourage gentle massage, wear gloves at night - Consider complimentary therapies: e.g., phototherapy (UVB) 3 times weekly for a 3-week trial; acupuncture. Very little evidence exists for these alternative therapies.	Topical - Capsaicin 0.025% or 0.03% ointment - Pramoxine 1% - Menthol/camphor/phenol – 0.3% each γ-Linolenic acid cream 2.2% Systemic First line: gabapentin (50–300 mg daily) 2–3 hours before sleep. Second line: tricyclic antidepressant such as doxepin 10 mg daily at night	Topical These agents can be applied two times daily (four times daily for capsaicin). Capsaicin may cause burning to the area initially. Menthol, camphor, and phenol are separate products that can be added to most creams. All three may be added together, commonly with a 0.3% concentration for each. Systemic The most common side effects of gabapentin are drowsiness, dizziness, confusion, fatigue, and occasionally peripheral edema. Potential adverse effects of tricyclic antidepressants include dizziness, blurred vision, constipation, and urinary retention. There is an increased risk of confusion and sedation, particularly in older adults.

Table 62.4 Symptom Management for Patients With Advanced Chronic Kidney Disease⁶⁷ (Cont'd)**Nausea and Vomiting**

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
- Metabolic disturbances (e.g., uremia) - Medications (e.g., opioids, SSRI antidepressants) - Gastrointestinal disturbances (e.g., constipation, delayed gastric emptying)	- Manage constipation - Encourage good oral hygiene - Smaller, more frequent meals; eat meals slowly - Avoid alcohol - Avoid foods that are greasy, spicy, or excessively sweet - Minimize aromas (e.g., cooking odors, perfumes, smoke) - Encourage relaxed, upright position after eating to facilitate digestion - Loose-fitting clothing - Consider complementary therapies (e.g., relaxation techniques, acupressure, the use of ginger)	First line: Ondansetron 4–8 mg every 8 hours as needed Second line: metoclopramide 2.5 mg every 4 hours as needed Third line: olanzapine 2.5 mg every 8 hours as needed OR haloperidol 0.5 mg every 8 hours as needed Fourth line: for persistent and severe nausea, consider increasing haloperidol to 1.0 mg (maximum 5 mg in 24 hours) OR replacing with methotrimeprazine 5 mg orally or 6.25 mg subcutaneously every 8 hours as needed	Ondansetron can be constipating. Haloperidol, metoclopramide, and olanzapine are all dopamine antagonists: avoid prescribing them together. They can also exacerbate RLS. They all cross the blood–brain barrier and extrapyramidal symptoms are possible. Haloperidol has a higher risk of extrapyramidal symptoms than metoclopramide and olanzapine. Increasing the dose of methotrimeprazine may lead to levels of drowsiness that the patient may find unacceptable and should be discussed with the patient and/or family.

Breathlessness

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
- Anxiety - Anemia - Infection - Volume overload leading to pulmonary edema	- Sit in an upright position (e.g., 45°) - Position by a window or use a fan to blow air gently across the face - Maintain a humid environment - Pursed lip breathing - Supplemental oxygen - Complementary therapies (e.g., relaxation techniques, music) - Consider role of diet such as sodium and fluid restriction if patient is volume overloaded	If patient is intravascularly volume overloaded: diuretic such as a loop diuretic—furosemide. Occasionally patients may require combination diuretic therapy—consider adding metolazone. Near the end of life: low doses of opioids are the most effective treatment. For breathlessness that is episodic and primarily associated with a specific activity, consider fentanyl 12.5 µg subcutaneously or sublingually PRN. For shortness of breath that is more constant or unpredictable in nature, consider hydromorphone 0.5 mg PO (0.2 mg subcutaneously) every 4 hours around the clock and every hour as needed.	Due to the accumulation of metabolites, opioids should always be started at a low dose and monitor closely for adverse effects. Due to its fast action, fentanyl works well in cases where breathlessness is predictable.

Continued on following page

Table 62.4 Symptom Management for Patients With Advanced Chronic Kidney Disease⁶⁷ (Cont'd)**Fatigue and Sleep Disturbances**

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
Fatigue • Vitamin D deficiency • Metabolic acidosis • Hyperphosphatemia • Secondary hypothyroidism • Anemia • Malnutrition • Mood disorders • Sleep disturbances Sleep Disturbances • Other symptoms (e.g., restless legs, pruritus, pain, breathlessness) • Cognitive impairment • Medications • Generalized insomnia • Mood disorders • Sleep apnea	Fatigue • Exercise • Nutrition and hydration management • Energy-conservation strategies • Good sleep hygiene (e.g., avoid stimulants before bed, avoid napping during the day, save the bedroom for sleep) • Cognitive and psychological approaches (e.g., relaxation therapy, delegating, and setting limits) • Complementary treatments (e.g., acupressure, massage, acupuncture)	Sleep Disturbances First line: consider low-dose gabapentin (50–300 mg at night), especially if the patient has concomitant neuropathic pain, RLS, or uremic pruritus Second line: Doxepin 10 mg at bedtime, especially if concomitant pruritus or neuropathic pain reported Third line: cautiously consider mirtazapine 7.5 mg or zopiclone 3.75–5 mg at night or melatonin 2–5 mg at night	Reassess medications after 2–4 weeks. Avoid over-the-counter sleep aids and benzodiazepines if possible. Specifically, avoid mirtazapine if taking tramadol or antidepressants. Monitor doxepin for anticholinergic side effects (e.g., dizziness, blurred vision, constipation, urinary retention, and cardiac arrhythmias). Evidence for melatonin is limited and inconclusive. Ideally, all of these medications should be prescribed for short-term use only.

Nociceptive Pain

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
• Determine cause for pain and consider appropriate investigations	• Physical therapies (e.g., physical therapy, aerobic exercise, stretching, massage, acupressure, acupuncture) • Behavioral therapies (e.g., cognitive behavioral therapy—most commonly used behavioral therapy), biofeedback, relaxation techniques, psychotherapy/individual or group counselling, guided imagery, mindfulness-based stress reduction • Interventional and surgical (e.g., ablative techniques, nerve blocks, trigger point injections)	Step 1: Acetaminophen/paracetamol, maximum of 3 g daily. If pain is localized to a small joint, consider a topical NSAID (e.g., diclofenac gel 5% or 10% two to three times daily). Step 2: There are no safe step 2 analgesics for nociceptive pain. Step 3: ADD a strong opioid to step 1 in very low doses. Hydromorphone starting at 0.5 mg PO (0.2 mg subcutaneously) every 4–6 hours; buprenorphine/fentanyl/methadone.	Trial each step for 1–4 weeks before progressing, depending on pain severity. Before starting a strong opioid, consider completing an opioid risk tool and order a bowel routine to avoid constipation (e.g., PEG 3350). All opioids should be started at low doses, monitored carefully for adverse effects and overall benefit, and titrated slowly (see Table 62.5).

Table 62.4 Symptom Management for Patients With Advanced Chronic Kidney Disease⁶⁷ (Cont'd)**Neuropathic Pain**

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
Determine cause for pain and consider appropriate investigations	As for nociceptive pain	Start with adjuvant therapy. First line: gabapentin, pregabalin (calcium channel alpha 2–delta ligands) Second line: tricyclic antidepressants, amitriptyline starting at 10–25 mg daily or doxepin starting at 10 mg daily. If additional analgesia is required in addition to adjuvant therapy, add a nonopioid and then proceed low dose of a strong opioid and titrate as described for nociceptive pain.	Methadone may be effective for severe neuropathic pain because of its activity against the NMDA receptor antagonism.

NMDA, N-methyl-D-aspartate; NSAID, nonsteroidal antiinflammatory drug; PO, per os; q4h, every 4 hours; PRN, as needed; RLS, restless legs syndrome; SSRI, selective serotonin reuptake inhibitor; UVB, short-wave ultraviolet B.

general principles are worth highlighting. Chronic pain can be defined as any painful condition that persists for more than 3 months.⁶⁹ Dialysis patients may also experience recurring episodes of acute pain, such as pain from needling fistulas or intradialytic steal syndrome, intradialytic headaches, and cramps. These acute pains tend to be associated with tissue damage but typically have no progressive pattern, last a predictable period, subside as healing occurs, and are episodic with periods without pain. By contrast, chronic pain is present for long periods and is often out of proportion with the extent of the originating injury. It is more likely to result in functional impairment and disability, psychological distress, sleep deprivation, and poor QOL.

Nonpharmacologic therapies have become a vital part of managing chronic pain and are often required to augment pharmacologic treatments to achieve adequate relief (i.e., multimodal therapy). They may also be used as stand-alone therapies. Examples are outlined briefly in Table 62.4. Core principles in developing a treatment plan for chronic pain include explaining the nature of the chronic pain condition, setting appropriate goals, and developing a comprehensive treatment approach and plan for adherence. Medication should not be the sole focus of treatment and should only be used when needed, in conjunction with other treatment modalities, to meet treatment goals. Optimal patient outcomes often require multiple approaches used in concert, coordinated via a multidisciplinary team.

There are five essential principles for the pharmacologic management of pain. These are described in Table 62.5. Of particular importance is the careful selection of analgesics when prescribing for patients with advanced CKD. Many analgesics and their metabolites are excreted by the kidney

through glomerular filtration, tubular secretion, or both and are at risk for accumulation of toxic metabolites if not monitored carefully. The choice of an appropriate initial therapeutic strategy is dependent on an accurate evaluation of the cause of the pain and the type of chronic pain syndrome. In particular, neuropathic pain should be distinguished from nociceptive pain (Table 62.6). For most patients with advanced CKD, the initial treatment of neuropathic pain involves calcium channel alpha 2–delta ligands (e.g., gabapentin and pregabalin) or a tricyclic antidepressant (see Table 62.4). Opioid medications are second-line agents for most patients with neuropathic pain. By contrast, the pharmacologic approach to nociceptive pain primarily involves nonnarcotic and opioid analgesics if nonpharmacologic strategies are insufficient. The strong opioids that are generally considered safer for use in patients with advanced CKD are hydromorphone, buprenorphine, fentanyl, and methadone, with doses started low, titrated slowly, and the patient monitored carefully for analgesia, adverse effects, overall function, and QOL (Table 62.4).

SPECIAL CONSIDERATIONS AT THE END OF LIFE

As a patient's condition deteriorates, certain nonpharmacologic interventions will become less realistic (e.g. exercise). Energy conservation and restoration will become of utmost importance. Ensure that appropriate supports are in place to assist with activities of daily living and that nursing care is available as needed. Drowsiness may increase as the end-of-life approaches due to disease progression (and/or

Table 62.5 Five Principles of Pain Management in Advanced CKD

Principle	Description	Specific Considerations in Advanced CKD
"By mouth"	- Oral administration is the safest and therefore preferred. - Patient comfort and effectiveness must be considered. If ingestion or absorption is uncertain, analgesics need to be given by alternative routes, such as transdermal, rectal, or subcutaneous.	Hemodialysis patients have easy intravenous access. However, this is to be avoided as the route of administration for analgesics to optimize safety and minimize the risk of abuse and addiction.
"By the clock"	For continuous or predictable pain, analgesics should be given regularly. Additional "breakthrough" or "rescue" medication should be available on an "as needed" (PRN) basis in addition to the regular dose.	Some patients with mild pain may achieve adequate pain relief with analgesic dosing posthemodialysis only. An example would be mild neuropathic pain dosed with gabapentin postdialysis.
"By the ladder"	Pharmacologic management proceeds stepwise from nonopioids (step 1) to very low doses of strong opioids (step 3) using the WHO analgesic ladder. There are no good options for step 2 analgesics for patients with advanced CKD.	- Careful selection of analgesics for each step of the ladder, taking into account degree of kidney failure, is critical (see Table 62.4). - Sustained-release preparations are generally not recommended in patients with advanced CKD.
"For the individual"	- There is large variability in patients' response to analgesics. The "correct" dose is the amount needed to relieve the pain without producing intolerable side effects. - Evaluation and recording of benefit and toxicity are essential.	Chronic pain is often experienced in the context of numerous other physical, psychosocial, and spiritual concerns, including end-of-life issues. Close attention to these other issues must not be forgotten as part of the pain management strategy.
"Attention to detail"	- Pain changes over time; therefore there is the need for ongoing reassessment. - Side effects should be explained and managed actively (e.g., constipation).	- There are no studies on the long-term use of analgesics in patients with CKD. Careful attention must be paid to efficacy and safety. - The impact on overall symptom burden, physical function, emotional state, cognition, and QOL should be assessed routinely.

CKD, Chronic kidney disease; QOL, quality of life; WHO, World Health Organization.

Table 62.6 Categories of Pain

	Neuropathic Pain	Nociceptive Pain
Description	Pain that results from damage to the nervous system resulting in either dysfunction or pathological change.	Pain that results from tissue damage in the skin, muscle, and other tissues, causing stimulation of sensory receptors.
Pain Characteristics	- Common descriptors include burning, tingling, and shooting. - May be associated with episodes of spontaneous pain, hyperalgesia, and allodynia. The presence of the latter is pathognomonic. - The pain may be felt at a site distant from its cause, in the distribution of a nerve, for example.	- Pain may be described as sharp, like a knife—often felt at the site of damage. - Pain may also be experienced as dull, aching, and poorly localized (as with stimulation of visceral nociceptors).
Response to Analgesics	Generally responds poorly to analgesics and requires the use of adjuvant therapy.	Generally responds well to analgesics.

medications). Some patients and families may even prefer increased sleepiness if the patient remains comfortable. Symptoms should be anticipated with the appropriate prescriptions in place to address a patient's symptoms as they appear, including an alternative route to oral as swallowing becomes compromised. [Fig. 62.3](#) outlines a terminal care algorithm for pain, which can also be used for breathlessness.⁶⁷ As a general rule, benzodiazepines should be avoided in patients with advanced CKD, especially those who are functionally impaired and/or more frail. Benzodiazepines can carry significant risks, including an increased risk of falls, fractures, and decreased cognition. However, they may be the only

option for a patient who can no longer swallow and may be beneficial for restless legs, anxiety, and agitation near the end of life. [Fig. 62.4](#) describes a terminal care approach to restlessness, agitation, and delirium.⁶⁷

ADVANCE CARE PLANNING

The unpredictable nature of CKD progression and the associated complications make it particularly important for patients and families to plan for end-of-life care needs.³⁵ Advanced care planning (ACP) is a process that involves

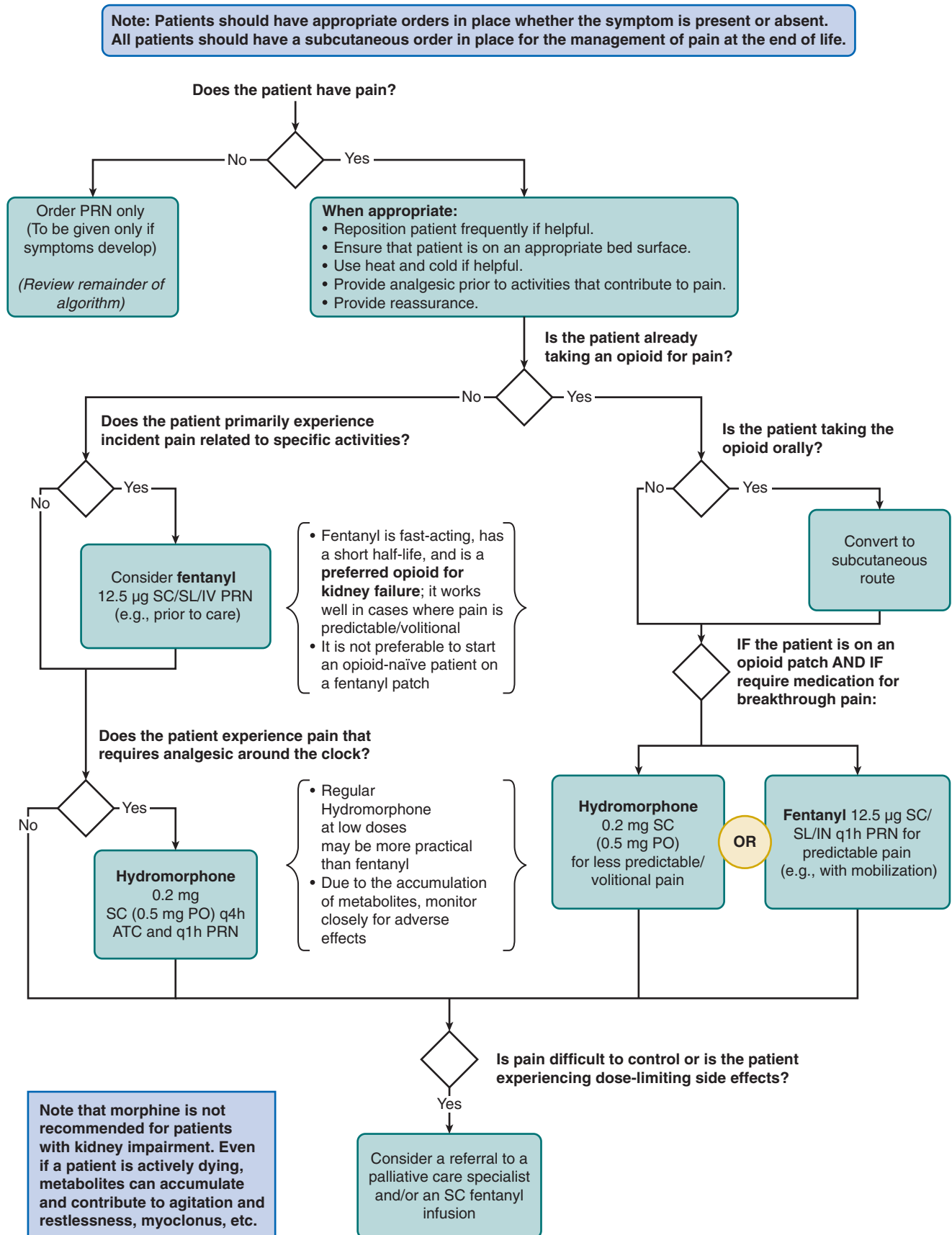


Fig. 62.3 Algorithm for the management of pain at the end of life for patients with advanced chronic kidney disease. ATC, Around the clock; PO, per os; q1h, every 1 hour; q4h, every 4 hours; SC, subcutaneous; SL, sublingual. (Adapted with permission from Sara N. Davison, Kidney Supportive Care Research Group. *Conservative Kidney Management Care Pathway*. 2016. Available from: www.ckmcare.com. Last accessed September 20, 2017.)

***All patients should have a subcutaneous/sublingual order for the management of restlessness at end of life whether the symptom is present or absent.**

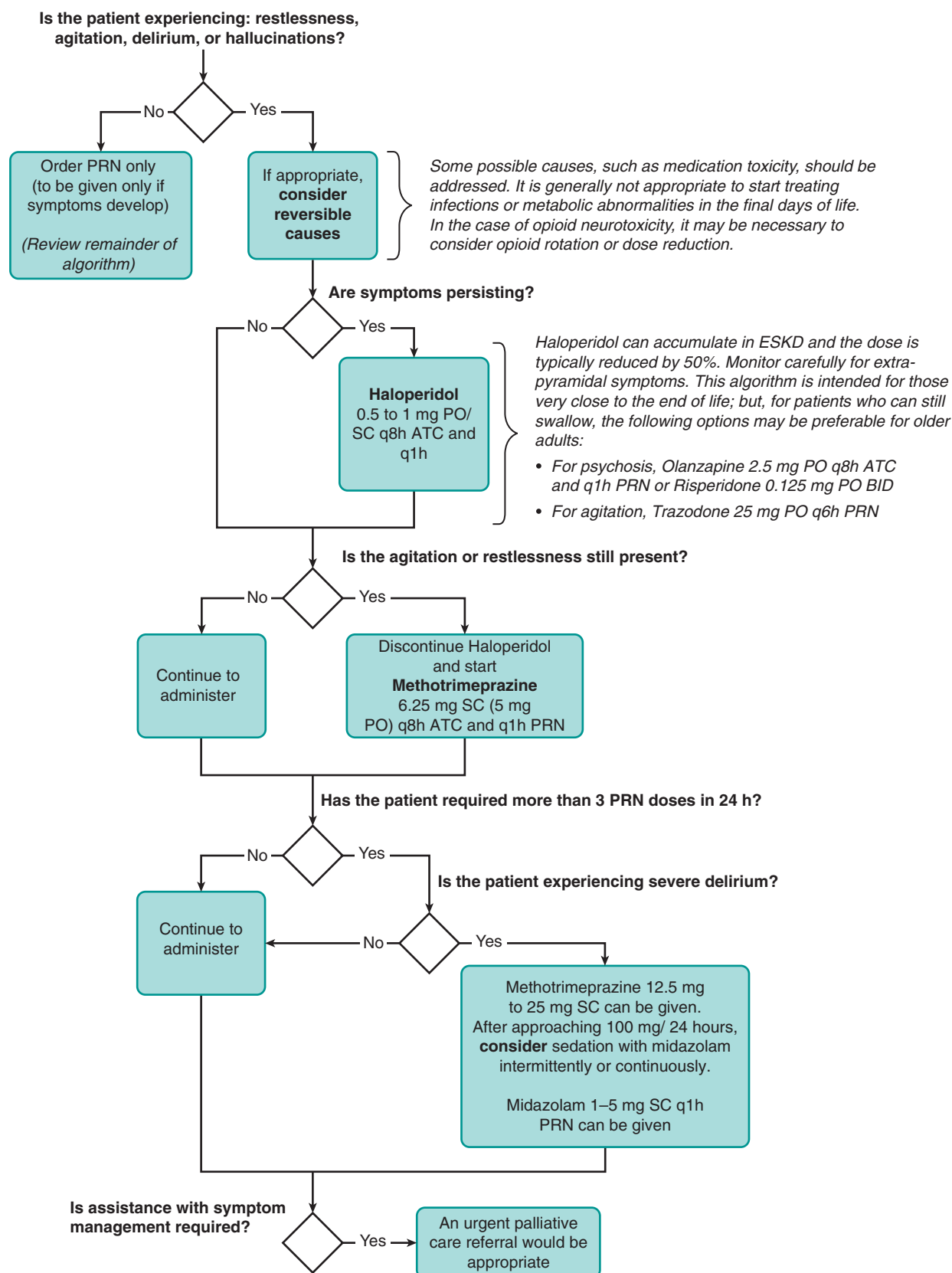


Fig. 62.4 Algorithm for the management of restlessness, agitation, and delirium at the end of life for patients with advanced chronic kidney disease (CKD). ATC, Around the clock; BID, twice a day; ESKD, end-stage kidney disease; PO, per os; PRN, as needed; q1h, every 1 hour; q6h, every 6 hours; q8h, every 8 hours; SC, subcutaneous. (Adapted with permission from Sara N. Davison, Kidney Supportive Care Research Group. *Conservative Kidney Management Care Pathway*. 2016. Available from: www.ckmcare.com. Last accessed September 20, 2017.)

understanding, communication, and discussion between a patient, the family (or other caregiver), and health-care providers, for the purpose of clarifying preferences for end-of-life care. It lays out a set of relationships, values, and processes for approaching end-of-life decisions for individual people when they can no longer speak for themselves, including attention to ethical, psychosocial, and spiritual issues relating to starting, continuing, withholding, and stopping dialysis.^{1,70} The unique circumstances around treatment options for patients with advanced CKD, such as palliative dialysis, CKM, and dialysis withdrawal, are discussed in greater

detail later and further perspectives are also discussed in Chapters 82 and 84. When goals for care have not been established, patients often experience unnecessary hospitalizations, unwanted invasive treatments, a prolonged dying process, and ultimately the type of death they would not have wanted.²

An approach to ACP for patients with advanced CKD was developed based on the perspectives of patients with advanced CKD^{70,71} and is illustrated in Fig. 62.5. ACP is a health behavior and these conversations need to be tailored to the readiness of the individual.⁷¹ Behavior change theories such as the

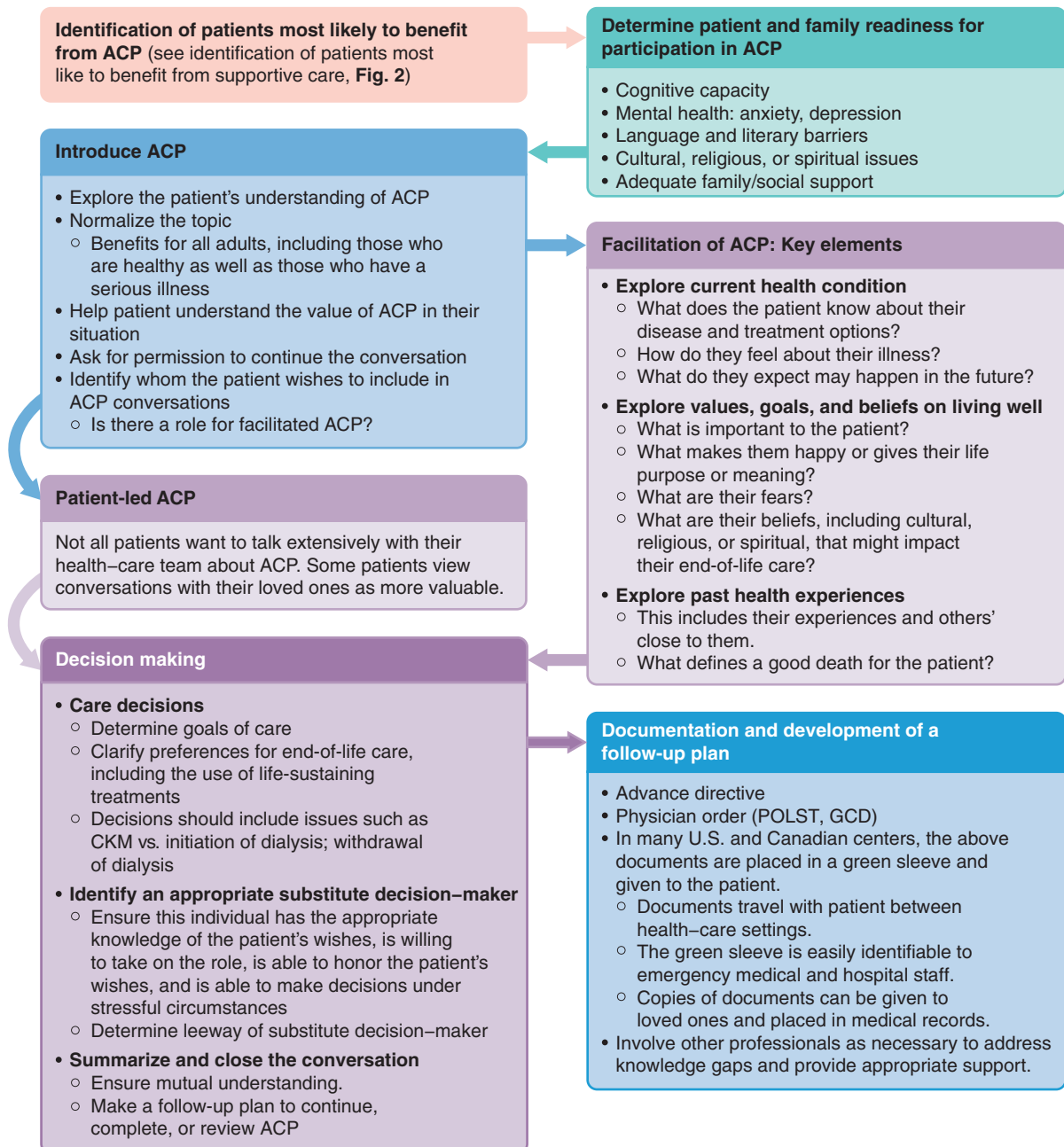


Fig. 62.5 A framework for facilitating advanced care planning (ACP) in patients with advanced chronic kidney disease. CKM, Conservative kidney management. (Adapted from Davison SN. Facilitating advance care planning for patients with end-stage renal disease: the patient perspective. *Clin J Am Soc Nephrol.* 2006;1:1023–1028; Davison SN, Torgunrud C. The creation of an advance care planning process for patients with ESRD. *Am J Kidney Dis.* 2007;49:27–36.)

trans-theoretical model or the “stages of change” model provide a useful framework for assessing patient readiness to engage in ACP.⁷² Patients go through the stages of pre-contemplation (no intention of engaging in ACP in the near future), to contemplation, to preparation, and finally to action. This framework takes into account an individual’s weighing of the pros and cons of engaging in ACP and assesses their attitudes about barriers to and facilitators of behavior change. A patient who is cognitively impaired or who is depressed or extremely anxious may have limited ability to participate meaningfully in ACP. The evidence suggests that a patient is much more likely to engage actively in ACP when he or she understands the personal relevance and perceived benefit.⁷¹ Cultural, spiritual, and religious beliefs and variances around the concepts of autonomy, decision-making authority, communicating prognosis, and views around care at the end of life may influence attitudes toward ACP.^{73–75} Clinicians should address any misconceptions the patient may have toward ACP and highlight the value of these conversations as it relates to that specific individual.

There are several key elements to ACP conversations that involve exploring the patient’s understanding of their illness, past health–care experiences, and values and beliefs (see Fig. 62.5). It is only through this exploration and understanding that the clinician will be able to align treatment options and end-of-life care with the patient’s personal goals for living and dying well. It is not necessary that all elements be addressed in a single conversation, but rather that over time both the patient and the clinician gain a full understanding of the patient’s beliefs, values, and overall goals for care. In routine clinical practice it is common for these conversations to build on each other in subsequent visits. Although many clinicians feel they lack the skills and confidence to have ACP conversations with their patients,³⁴ effective ACP communication skills can be learned,^{76,77} especially when using the strategies for culturally sensitive shared decision making outlined in Table 62.1.

Ideally, ACP should be integrated into routine kidney care early and readdressed throughout the patient’s illness. This helps to normalize the process; to afford patients the time to think, reflect, and make well-informed choices for future care; and to ensure care plans remain aligned with their preferences and prognosis as their health state declines. This often proves challenging as patients may feel that ACP is not yet relevant for them early in their disease. However, sudden illness or complications can occur without warning. At a minimum, however, high-risk patients should be identified through regular supportive care assessments as outlined in Fig. 62.2, to facilitate timely conversations with those most in need.

CONSERVATIVE KIDNEY MANAGEMENT

There is growing recognition that dialysis is unlikely to benefit and may even harm some patients, particularly those with significant preexisting functional or cognitive impairment and high levels of comorbidity.^{78–81} Frail patients often experience accelerated functional and cognitive decline after starting dialysis.^{82,83} Patients residing in long-term care facilities at the time of starting dialysis have especially poor outcomes.⁸² CKM is an appropriate alternative to chronic

dialysis for patients unlikely to benefit from dialysis (see also Chapter 84).

CKM can be defined as holistic patient-centered care for patients with G5 CKD that through shared decision making emphasizes QOL, active symptom management, and advance care planning. It includes interventions to delay progression and minimize complications of CKD but it does not include dialysis.¹ Some elderly patients managed conservatively live as long as patients who choose to start dialysis with better preservation of function, fewer admissions to acute care settings, while avoiding the burdens of dialysis and remaining in their home communities, resulting in a better QOL.^{78–81} The international nephrology community recognizes that for patients unlikely to benefit from dialysis, dialysis should not be the default therapy and advocates strongly for the provision of quality CKM.¹

Choosing CKM does not mean imminent death. Many patients live for months or years with G5 CKD. Nor does CKM imply a “no care” philosophy. In fact, CKM requires highly integrated communication and care that involves tailor-made combinations of structures, processes, and interventions to address unique patient needs and unique system-community circumstances. Careful planning is necessary to link and transition patients at various points along the continuum of care, such as between primary, secondary, and tertiary care; between community-based and institutional care; and between acute care and long-term care. It may require involvement of mental health–care and social services and should assist with referral to specialist supportive/palliative care and hospice services as available and appropriate. Multidisciplinary care teams will be required to deliver CKM. How best to implement CKM throughout complex, multisector health–care systems is unclear. One method of implementation is through a clinical pathway that operationalizes best-practice guidelines into “how to” steps for care delivery. A successful example of this is the interactive, online Canadian CKM pathway that includes an integrated plan of communication and care and a CKM patient decision aid that has been implemented across the province of Alberta and has shown to be effective in both increasing uptake of evidence-based care and optimizing patient management and outcomes (Fig. 62.6).⁶⁷

Until recently, there were no clear recommendations for managing the metabolic complications of CKD. Current CKD guidelines were established for optimizing long-term health outcomes and do not address the priorities of care for patients who have chosen CKM. CKM-specific recommendations for both CKD management and symptom control (as outlined in Table 62.4) have recently been suggested to assist with the standardization and evaluation of high-quality CKM care.⁶⁸ These recommendations and their associated rationales are outlined in Table 62.7. The illness trajectory of patients on a CKM pathway can be highly variable and is anticipated to pass through several stages. Patients with an estimated glomerular filtration rate (eGFR) of 10–15 mL/min/1.73 m² may remain relatively functional and stable for many months to years, whereas a patient who spends much of their time lying down with an eGFR of 5 mL/min/1.73 m² likely has a life expectancy measured in weeks to a couple of months. Comprehensive CKM guidelines need to take the patient’s general condition, prognosis, and goals of care into account. Earlier in the disease trajectory, maximizing QOL likely

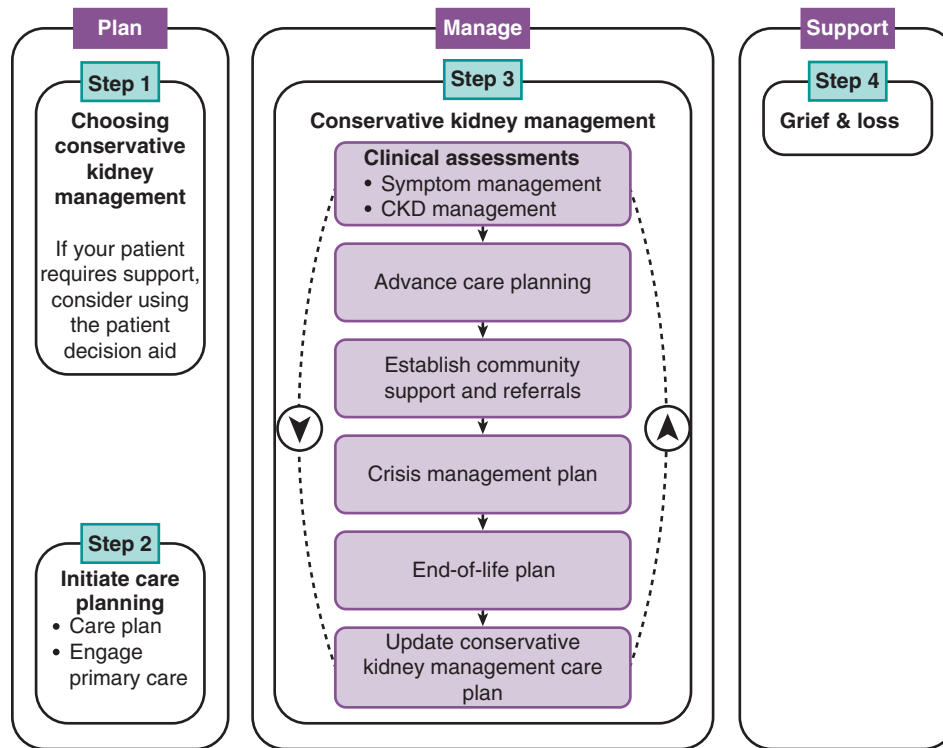


Fig. 62.6 An integrated approach to the care of patients who have chosen conservative kidney management.⁶⁷ CKM, Conservative kidney management. (Reproduced with permission from Dr. Sara Davison, Kidney Supportive Care Research Group.)

Table 62.7 Summary of CKD Management Recommendations for Patients Who Have Chosen CKM^{67,68}

Guideline	Treatment Rationale	Recommended Interventions
Dyslipidemia	Patients are unlikely to benefit from treating dyslipidemia in the last few years of life. Patients may gain improvement in QOL from stopping statin medications. ^{103,104}	Care providers, in discussion with the patient, can discontinue statin medications.
Blood Pressure	The primary goal of blood pressure management is to optimize physical and cognitive function and minimize the risk of falls, while avoiding very high readings. Decisions about specific medications would depend on the patient's comorbidities. Diuretics are a unique consideration and are aimed primarily at the treatment of volume overload that causes breathlessness or symptomatic peripheral edema.	Blood pressure targets can be relaxed for most CKM patients to $\leq 160/90$ mm Hg. ^{104–106} This applies to patients with diabetes as well.
Sodium Restriction	High sodium intake may be associated with volume overload leading to breathlessness and symptomatic peripheral edema. Because sodium restriction can influence palatability, strategies should balance symptom management and the patient's priorities for care and other issues such as appropriate nutrition.	Diuretics and dietary sodium restriction to assist with volume control if volume overload is contributing to symptom burden.
Anemia	Anemia can contribute to fatigue and breathlessness. The purpose of treating anemia would be to reduce these symptoms as opposed to reducing cardiac mortality or morbidity. When a patient starts spending most of his or her time lying down and/or the end of life is near, it is no longer appropriate to manage fatigue and breathlessness by addressing anemia and anemia treatment can be stopped.	Erythropoiesis-stimulating agents and iron supplementation. Anemia-related bloodwork every 3–6 months is appropriate but should be based on patient preference and symptoms. It is appropriate to stop managing anemia in the last weeks or days of life.
Hyperkalemia	Hyperkalemia predisposes patients to cardiac arrhythmias and sudden death. Acute treatment of hyperkalemia is appropriate if consistent with the patient's goals. If patients wish to liberalize their potassium intake, the risks of lifting the restriction must be explained clearly.	Interventions include a potassium-restricted diet and the use of potassium-binding resins such as sodium polystyrene sulfonate. For those aiming to maintain normal potassium levels, it is reasonable to monitor potassium levels monthly. It is appropriate to stop monitoring and managing potassium levels in the last weeks or days of life.

Continued on following page

Table 62.7 Summary of CKD Management Recommendations for Patients Who Have Chosen CKM^{67,68}
(Cont'd)

Guideline	Treatment Rationale	Recommended Interventions
Acidosis	Metabolic acidosis can contribute to fatigue, bone loss, and muscle wasting ¹⁰⁷ ; Treatment is aimed at managing these symptoms. If the patient finds the pill burden too great, or when the patient can no longer swallow, treatment should be stopped.	The primary intervention is the use of sodium bicarbonate. It is reasonable to monitor bicarbonate every 3–6 months if patients are being treated. It is appropriate to stop monitoring and managing bicarbonate levels in the last weeks or days of life.
Calcium/ Phosphorus	We recommend promoting QOL through liberalizing diet and maintaining adequate nutrition rather than normalizing biochemistry as it is not clear that there are benefits to normalizing in patients being cared for conservatively in the last few years of life. Rather, there is a possibility of harm in promoting lower protein intake in patients already at high risk for protein malnutrition. Hyperphosphatemia may contribute to RLS; calcium and phosphorous depositions can lead to myalgias, arthralgias, and pseudogout. Interventions to partially normalize biochemistry should only be considered to minimize these symptoms.	If the patient is experiencing symptoms, interventions include a phosphorus-restricted diet (being careful to maintain adequate nutrition) and the use of phosphate binders such as calcium carbonate. Bloodwork every 3–6 months is appropriate if patients are being treated but should be based on patient preference and symptoms.
Vitamin D	Vitamin D may have a role in the symptoms of fatigue, weakness, and muscle loss.	Low-dose active vitamin D (vitamin D analog) may be beneficial. There is no additional benefit in monitoring parathyroid hormone levels.

CKD, Chronic kidney disease; CKM, conservative kidney management; QOL, quality of life; RLS, restless legs syndrome.

requires a careful balance between promoting function and addressing symptom burden, whereas comfort and control of symptoms generally take precedence in the last weeks and days of life. Although there is minimal level I–III evidence specific to CKM patients, these recommendations have incorporated both geriatric and palliative care principles and serve as a starting point for standardization of care that will need to be refined as evidence for best practice accumulates. Many CKM patients present with complex health scenarios and recommendations have to be individualized. By explaining the rationale and possible benefits and drawbacks of interventions, patients can become active participants in the shared decision-making process, ensuring that all interventions remain aligned appropriately with their prognosis, values, and preferences.

Shared decision making for CKM versus dialysis is not integrated routinely into practice and patients continue to enter dialysis care pathways that are aligned poorly with their preferences and prognosis. The substantial regional variations in rates of kidney failure treated with dialysis for older patients suggest that decisions to start dialysis are influenced by factors other than clinical outcomes and may be more reflective of physician preferences.⁸⁴ Recent studies report that most elderly patients feel they have no choice in the decision to start dialysis; are not informed fully of their options or the risks of dialysis; experience distress over unexpected complications and burdens of dialysis; desire more conversation about the decision; and rarely recall being given the option for CKM.^{85,86} Many patients feel a sense of resignation to dialysis, especially when decisions are framed incorrectly as choosing between life and death.³⁷ Physicians conceded that they struggle to explain illness trajectory and complexity and do not discuss

prognosis unless prompted. There are currently extremely limited comparative prognostic data on CKM and dialysis. However, a recent systematic review of survival of CKM versus dialysis for elderly patients demonstrated a broadly similar 1-year survival.⁸⁰ Patients who choose CKM tend to have stable symptom burden and QOL until the last 1 or 2 months of life, and symptoms are amenable to supportive care interventions.⁸¹ Given the degree of uncertainty about benefits and risks, and the potentially profound impact on QOL, patient values must be paramount in the decision about whether or not to have dialysis.

There has been great interest in the development of patient decision aids to help with renal replacement therapy decisions, including CKM. Patient decision aids are tools designed to facilitate people's ability to make reasoned decisions about care, which are compatible with their goals, values, and prognosis. These tools help supplement and guide the shared decision-making process and can be developed in multiple media forms, such as written materials, audiotapes, videos, Web-based formats, and oral presentations.⁸⁷ Two Cochrane reviews evaluating the use of patient decision aids have shown them to be effective^{52,88} and can improve peoples' knowledge regarding options, reduce their decisional conflict, improve understanding of the relative risks and benefits, increase their participation in decision making, and improve values-choice concordance.⁸⁹ However, most decision aids for dialysis treatment options either do not mention CKM or address it only in a peripheral way. The main decision support is around which form of dialysis (or transplantation) to choose. These decision aids also do not include patient-specific risks and benefits so are not positioned to weigh QOL versus survival advantage, which patients require to make the decision that

is best for them. New CKM decision aids are currently under development and evaluation to address these deficits.^{67,90} These tools, however, must be used with caution. They are educational tools only and should not be used to mandate certain care pathways over others.

PALLIATIVE DIALYSIS

As disease progresses, patients' goals of care may shift to focus almost exclusively on QOL rather than survival, with a strong emphasis on symptom control; advance care planning; and emotional, social, and family support. Palliative dialysis is an approach to dialysis that "prioritizes" QOL over survival and requires full integration of kidney supportive care (Fig. 62.7). Palliative dialysis is often, incorrectly, perceived as being equivalent to less dialysis or a precursor to withdrawal of dialysis. Although adaptation of dialysis timings or eventual withdrawal from dialysis may be a component, this alone will rarely ameliorate symptoms or suffering for patients.²⁵ This care pathway has recently been explored in greater detail elsewhere.²⁵ Dialysis patients may transition to a palliative approach to their dialysis care near the end of their life (see Chapter 84). Patients also may start dialysis with this palliative approach to their care. The dialysis guidelines and standards of care will need to permit flexibility so that care can be tailored appropriately for each patient and will look more like the CKM guidelines in Fig. 62.6.

WITHDRAWAL OF DIALYSIS

Withdrawal from dialysis remains a leading cause of death for dialysis patients, although there are significant differences in reported rates across studies and national renal registries⁹¹

and in practice patterns between countries.⁹² Likely many of these differences relate to variations in culture, policy and physician perceptions, and attitudes toward withdrawal. For most cultures, withdrawal of dialysis is ethically and clinically acceptable, when based on shared decision making that balances the patient's interests with local cultural norms.¹

Potential situations in which it is appropriate to withdraw dialysis are described in Box 62.1. It is incumbent upon all providers caring for a patient contemplating stopping dialysis to address potentially remedial factors contributing to the decision, such as depression or other symptoms such as pain, as well as potentially reversible social factors (discussed further in Chapter 82). Ensuring access to appropriate supportive care is an integral part of the care following a decision to withdraw dialysis.

Box 62.1 Situations in Which It Is Appropriate to Consider Withdrawal from Dialysis^{1,23}

1. Patients with decision-making capacity, who being fully informed and making voluntary choices, refuse dialysis or request that dialysis be discontinued.
2. Patients who no longer possess decision-making capacity who have previously indicated refusal of dialysis through appropriate advance care planning.
3. Patients who no longer possess decision-making capacity and whose properly appointed legal agents/surrogates refuse dialysis or request that it be discontinued.
4. Patients with irreversible, profound neurologic impairment such that they lack signs of thought, sensation, purposeful behavior, and awareness of self and environment.

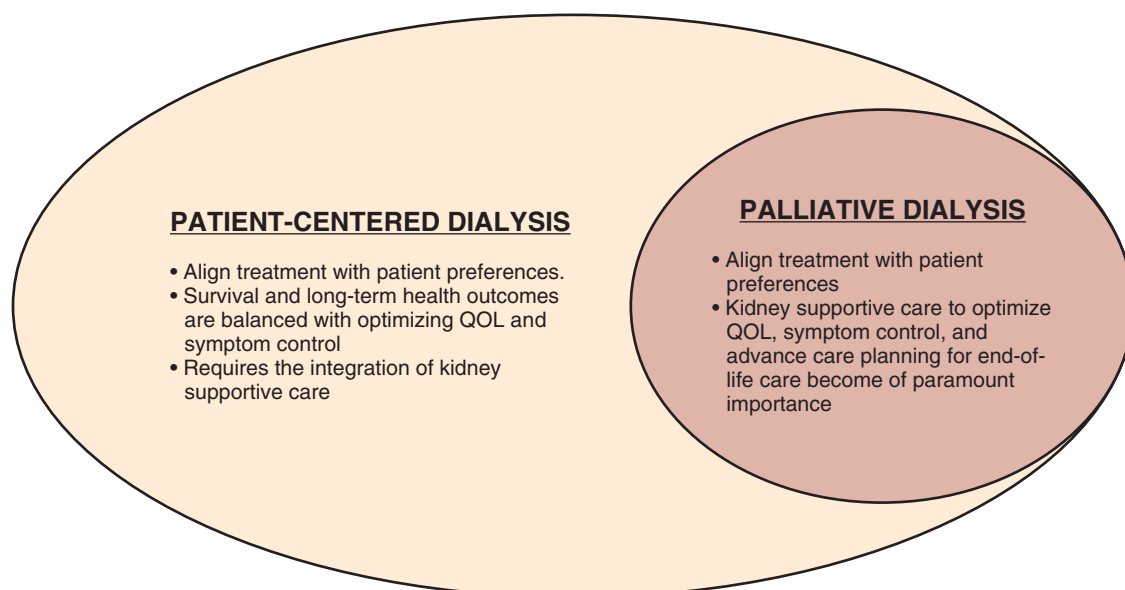


Fig. 62.7 Conceptualizing palliative dialysis. QOL, Quality of life. (Adapted from Davison SN, Jassal SV. Supportive care: integration of patient-centered kidney care to manage symptoms and geriatric syndromes. *Clin J Am Soc Nephrol*. 2016;11:1882–1891.)

EARLY IDENTIFICATION OF PATIENTS MOST LIKELY TO BENEFIT FROM SUPPORTIVE CARE SERVICES

A systematic approach to identify patients most likely to benefit from supportive care is critical in integrating kidney supportive care. Patients need to be identified based on need, not just poor prognosis. Not all patients require ongoing supportive care services, and requirements may fluctuate over the course of their illness trajectory (Fig. 62.1). However, most patients transition to a trajectory of progressive functional decline and/or experience complex clusters of physical and psychological symptoms. Early recognition of needs helps with planning, results in fewer hospital admissions and crises as patients approach the end of life, avoids needless suffering, and ultimately improves both patient and health system outcomes. Once the key elements of kidney supportive care are understood, it becomes clearer how to screen for patients in need. Fig. 62.2 highlights key issues that should be considered when screening patients for supportive care needs. At a minimum, this would include patients at a high risk of early death, those whose condition is deteriorating, patients with a high symptom burden or poor QOL, and those who are struggling with decision making and goals of care.

SUMMARY

Addressing supportive care needs is a high priority for patients with advanced CKD.²⁴ The international kidney community recognizes that access to supportive care services should be available to all patients with advanced CKD, although the extent and form of these services will depend on local resources. For optimal delivery, multidisciplinary teams will need to work collaboratively across health-care sectors. There will also need to be an increased emphasis on education in kidney supportive care that recognizes this as a core competency in nephrology. Integrating kidney supportive care will also require recognition that our current models of care, policies, and clinical standards do not consistently meet the needs of our patients. Ongoing work will help determine best practices, including appropriate quality of kidney supportive care metrics.

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

KEY REFERENCES

1. Davison SN, Levin A, Moss AH, et al. Executive summary of the KDIGO Controversies Conference on Supportive Care in Chronic Kidney Disease: developing a roadmap to improving quality care. *Kidney Int*. 2015;88:447–459.
2. Davison SN. End-of-life care preferences and needs: perceptions of patients with chronic kidney disease. *Clin J Am Soc Nephrol*. 2010;5:195–204.
3. Tong A, Cheung KL, Nair SS, et al. Thematic synthesis of qualitative studies on patient and caregiver perspectives on end-of-life care in CKD. *Am J Kidney Dis*. 2014;63:913–927.
4. Quill TE. Perspectives on care at the close of life. Initiating end-of-life discussions with seriously ill patients: addressing the “elephant in the room.” *JAMA*. 2000;284:2502–2507.
5. Combs SA, Culp S, Matlock DD, et al. Update on end-of-life care training during nephrology fellowship: a cross-sectional national survey of fellows. *Am J Kidney Dis*. 2014.
6. Culp S, Lupu D, Arenella C, et al. Unmet supportive care needs in U.S. dialysis centers and lack of knowledge of available resources to address them. *J Pain Symptom Manage*. 2015;51:756–761.
7. Davison S, Jhangri GS. Existential and supportive care needs among patients with chronic kidney disease. *J Pain Symptom Manage*. 2010;40:838–843.
8. Organization WH. WHO definition of palliative care. World Health Organization; 2008. <http://www.who.int/cancer/palliative/definition/en/>. Accessed April 5, 2016.
9. Davison SN, Jhangri GS, Koffman J. Knowledge of and attitudes towards palliative care and hospice services among patients with advanced chronic kidney disease. *BMJ Support Palliat Care*. 2014;6:66–74.
10. Brown EABH, Davison SN, Koffman J, et al. Supportive care: religious and multicultural aspects of advance care planning and shared decision-making including the role for decision science. *Clin J Am Soc Nephrol*. 2016;11:1915–1920.
11. Renal Physicians Association. In: Association, ed. Shared decision-making in the appropriate initiation of and withdrawal from dialysis. Renal Physicians Association and American Society of Nephrology; 2010. <https://www.renaldmd.org/catalogue-item.aspx?id=682>. Accessed April 5, 2016.
12. Manns B, Hemmelgarn B, Lillie E, et al. Setting research priorities for patients on or nearing dialysis. *Clin J Am Soc Nephrol*. 2014;9:1813–1821.
13. Davison SN, Jassal SV. Supportive care: integration of patient-centered kidney care to manage symptoms and geriatric syndromes. *Clin J Am Soc Nephrol*. 2016.
14. Brown EA, Bekker HL, Davison SN, et al. Supportive care: communication strategies to improve cultural competence in shared decision making. *Clin J Am Soc Nephrol*. 2016.
15. Kagawa-Singer M, Kassim-Lakha S. A strategy to reduce cross-cultural miscommunication and increase the likelihood of improving health outcomes. *Acad Med*. 2003;78:577–587.
16. Chiu HH, Tangri N, Djurdjev O, et al. Perceptions of prognostic risks in chronic kidney disease: a national survey. *Can J Kidney Health Dis*. 2015;2:53.
17. Schell JO, Patel UD, Steinhilber KE, et al. Discussions of the kidney disease trajectory by elderly patients and nephrologists: a qualitative study. *Am J Kidney Dis*. 2012;59:495–503.
18. Couchoud C, Hemmelgarn B, Kotanko P, et al. Supportive care: time to change our prognostic tools and their use in CKD. *Clin J Am Soc Nephrol*. 2016.
19. Walker SR, Gill K, Macdonald K, et al. Association of frailty and physical function in patients with non-dialysis CKD: a systematic review. *BMC Nephrol*. 2013;14:228.
20. Johnstone S, Dombro L, Garza GSCK, et al. Declines in hemodialysis patients’ physical and mental component scores before death. *J Am Soc Nephrol*. 2014;25.
21. Thijssen S, Wong MM, Usvyat LA, et al. Nutritional competence and resilience among hemodialysis patients in the setting of dialysis initiation and hospitalization. *Clin J Am Soc Nephrol*. 2015;10:1593–1601.
22. Kotanko P, Kooman J, van der Sande F, et al. Accelerated or out of control: the final months on dialysis. *J Ren Nutr*. 2014;24:357–363.
23. Stacey D, Legare F, Col NF, et al. Decision aids for people facing health treatment or screening decisions. *Cochrane Database Syst Rev*. 2014;(1):CD001431.
24. Davison SN, Jhangri GS, Johnson JA. Cross-sectional validity of a modified Edmonton symptom assessment system in dialysis patients: a simple assessment of symptom burden. *Kidney Int*. 2006;69:1621–1625.
25. Davison SN, Jhangri GS, Johnson JA. Longitudinal validation of a modified Edmonton symptom assessment system (ESAS) in haemodialysis patients. *Nephrol Dial Transplant*. 2006;21:3189–3195.
26. Davison SN, Jhangri GS. Impact of pain and symptom burden on the health-related quality of life of hemodialysis patients. *J Pain Symptom Manage*. 2010;39:477–485.
27. Edmonton Zone Palliative Care Program and Northern Alberta Renal Program. 2016. <http://www.palliative.org/tools.html>. Accessed March 31, 2018.
28. Davison SN, Koncicki H, Brennan F. Pain in chronic kidney disease: a scoping review. *Semin Dial*. 2014;27:188–204.
29. Dworkin RH, Turk DC, Wyrwich KW, et al. Interpreting the clinical importance of treatment outcomes in chronic pain clinical trials: IMMPACT recommendations. *J Pain*. 2008;9:105–121.

60. Andrew Moore R, Eccleston C, Derry S, et al. "Evidence" in chronic pain—establishing best practice in the reporting of systematic reviews. *Pain*. 2010;150:386–389.
63. Antunes B, Harding R, Higginson IJ. Implementing patient-reported outcome measures in palliative care clinical practice: a systematic review of facilitators and barriers. *Palliat Med*. 2014;28:158–175.
67. Kidney Supportive Care Research Group. *Conservative Kidney Management Care Pathway*, 2016. www.ckmcare.com. Accessed September 20, 2017.
68. Davison SN, Tupala B, Wasylynuk BA, et al. *Recommendations for the care of patients receiving conservative kidney management: focus on management of chronic kidney disease and symptoms*. *Clin J Am Soc Nephrol*. 2018 (in press).
71. Davison SN, Torgunrud C. The creation of an advance care planning process for patients with ESRD. *Am J Kidney Dis*. 2007;49:27–36.
74. Chambers EJ, Brown E, Germain M. *Supportive Care for the Renal Patient*. Oxford University Press; 2010.
77. Schell JO, Green JA, Tulsy JA, et al. Communication skills training for dialysis decision-making and end-of-life care in nephrology. *Clin J Am Soc Nephrol*. 2013;8:675–680.
80. Foote C, Kotwal S, Gallagher M, et al. Survival outcomes of supportive care versus dialysis therapies for elderly patients with end-stage kidney disease: a systematic review and meta-analysis. *Nephrology (Carlton)*. 2016;21:241–253.
81. Murtagh FE, Burns A, Moranne O, et al. Supportive care: comprehensive conservative care in end-stage kidney disease. *Clin J Am Soc Nephrol*. 2016.
82. Tamura MK, Covinsky K, Chertow G, et al. Functional status of elderly adults before and after initiation of dialysis. *N Engl J Med*. 2009;361:1539–1547.
83. Jassal SV, Chiu E, Hladunewich M. Loss of independence in patients starting dialysis at 80 years of age or older. *N Engl J Med*. 2009;361:1612–1613.
85. Ladin K, Lin N, Hahn E, et al. Engagement in decision-making and patient satisfaction: a qualitative study of older patients' perceptions of dialysis initiation and modality decisions. *Nephrol Dial Transplant*. 2016.
86. Song MK, Lin FC, Gilet CA, et al. Patient perspectives on informed decision-making surrounding dialysis initiation. *Nephrol Dial Transplant*. 2013;28:2815–2823.
90. Davis JL, Davison SN. Hard choices, better outcomes: a review of shared decision-making and patient decision aids around dialysis initiation and conservative kidney management. *Curr Opin Nephrol Hypertens*. 2017;26:205–213.
91. Murphy E, Germain MJ, Cairns H, et al. International variation in classification of dialysis withdrawal: a systematic review. *Nephrol Dial Transplant*. 2014;29:625–635.
94. Tangri N, Grams ME, Levey AS, et al. Multinational assessment of accuracy of equations for predicting risk of kidney failure: a meta-analysis. *JAMA*. 2016;315:164–174.
95. Thamer M, Kaufman JS, Zhang Y, et al. Predicting early death among elderly dialysis patients: development and validation of a risk score to assist shared decision making for dialysis initiation. *Am J Kidney Dis*. 2015;66:1024–1032.
99. Cohen LM, Ruthazer R, Moss AH, et al. Predicting six-month mortality for patients who are on maintenance hemodialysis. *Clin J Am Soc Nephrol*. 2010;5:72–79.
102. Dworkin RH, Turk DC, Farrar JT, et al. Core outcome measures for chronic pain clinical trials: IMMPACT recommendations. *Pain*. 2005;113:9–19.
103. Kutner JS, Blatchford PJ, Taylor DH Jr, et al. Safety and benefit of discontinuing statin therapy in the setting of advanced, life-limiting illness: a randomized clinical trial. *JAMA Intern Med*. 2015;175:691–700.
106. Musini VM, Tejani AM, Bassett K, et al. Pharmacotherapy for hypertension in the elderly. *Cochrane Database Syst Rev*. 2009;CD000028.

REFERENCES

1. Davison SN, Levin A, Moss AH, et al. Executive summary of the KDIGO Controversies Conference on Supportive Care in Chronic Kidney Disease: developing a roadmap to improving quality care. *Kidney Int.* 2015;88:447–459.
2. Wong SP, Kreuter W, O'Hare AM. Treatment intensity at the end of life in older adults receiving long-term dialysis. *Arch Intern Med.* 2012;172:661–663, discussion 663–664.
3. Ahmed N, Bestall JC, Ahmedzai SH, et al. Systematic review of the problems and issues of accessing specialist palliative care by patients, carers and health and social care professionals. *Palliat Med.* 2004;18:525–542.
4. Murray AM, Arko C, Chen SC, et al. Use of hospice in the United States dialysis population. *Clin J Am Soc Nephrol.* 2006;1:1248–1255.
5. Canadian Institute for Health Information. *Health Care Use at the End of Life in Western Canada*. Ottawa, ON: Canadian Institute for Health Information; 2007:1. electronic text (xi, 109 p)
6. The National Council for Palliative Care. *National Survey of Patient Activity Data for Specialist Palliative Care Services - MDS Full Report for the Year 2011-2012*. London, UK: The National Council for Palliative Care; 2011.
7. Davison SN. End-of-life care preferences and needs: perceptions of patients with chronic kidney disease. *Clin J Am Soc Nephrol.* 2010;5:195–204.
8. Tong A, Cheung KL, Nair SS, et al. Thematic synthesis of qualitative studies on patient and caregiver perspectives on end-of-life care in CKD. *Am J Kidney Dis.* 2014;63:913–927.
9. Quill TE. Perspectives on care at the close of life. Initiating end-of-life discussions with seriously ill patients: addressing the “elephant in the room”. *JAMA.* 2000;284:2502–2507.
10. Combs SA, Culp S, Matlock DD, et al. Update on end-of-life care training during nephrology fellowship: a cross-sectional national survey of fellows. *Am J Kidney Dis.* 2014.
11. Culp S, Lupu D, Arenella C, et al. Unmet supportive care needs in U.S. dialysis centers and lack of knowledge of available resources to address them. *J Pain Symptom Manage.* 2015;51:756–761.
12. Davison S, Jhangri GS. Existential and supportive care needs among patients with chronic kidney disease. *J Pain Symptom Manage.* 2010;40:838–843.
13. Organization WH. *WHO Definition of Palliative Care*. World Health Organization; 2008. <http://www.who.int/cancer/palliative/definition/en/>. Accessed April 5, 2016.
14. Payne S, Sheldon F, Jarrett N, et al. Differences in understanding of specialist palliative care amongst service providers and commissioners in South London. *Palliat Med.* 2002;16:395–402.
15. Davison SN, Jhangri GS, Koffman J. Knowledge of and attitudes towards palliative care and hospice services among patients with advanced chronic kidney disease. *BMJ Support Palliat Care.* 2014;6:66–74.
16. Dalal S, Palla S, Hui D, et al. Association between a name change from palliative to supportive care and the timing of patient referrals at a comprehensive cancer center. *Oncologist.* 2011;16:105–111.
17. *Public Opinion Research on Palliative Care—A Report Based on Research by Public Opinion Strategies*. New York: Center to Advance Palliative Care; 2011.
18. Wentlandt K, Krzyzanowska MK, Swami N, et al. Referral practices of oncologists to specialized palliative care. *J Clin Oncol.* 2012;30:4380–4386.
19. Institute P-COR. *Patient-Centered Outcomes Research*. Patient-Centered Outcomes Research Institute; 2012. <http://www.pcori.org/>. Accessed April 5, 2016.
20. Research SIP-O. *Strategy for Patient-Oriented Research*. Canadian Institutes of Health Research; 2013. <http://www.cihr-irsc.gc.ca/e/41204.html>. Accessed April 5, 2016.
21. INVOLVE. *National Institute for Health Research*; 1996. <http://www.invo.org.uk/about-involve/>. Accessed April 5, 2016.
22. Brown EABH, Davison SN, Koffman J, et al. *Supportive Care: Religious and multicultural aspects of advance care planning and shared decision-making including the role for decision science C/ASN*; 2016.
23. Renal Physicians Association. In: Association, R. P., ed. *Shared Decision-Making in the Appropriate Initiation of and Withdrawal From Dialysis*. Renal Physicians Association and American Society of Nephrology; 2010. <https://www.renalmnd.org/catalogue-item.aspx?id=682>. Accessed April 5, 2016.
24. Manns B, Hemmelgarn B, Lillie E, et al. Setting research priorities for patients on or nearing dialysis. *Clin J Am Soc Nephrol.* 2014;9:1813–1821.
25. Davison SN, Jassal SV. Supportive care: integration of patient-centered kidney care to manage symptoms and geriatric syndromes. *Clin J Am Soc Nephrol.* 2016.
26. Davison SN. The ethics of end-of-life care for patients with ESRD. *Clin J Am Soc Nephrol.* 2012;7:2049–2057.
27. Baker ME. Economic, political and ethnic influences on end-of-life decision-making: a decade in review. *J Health Soc Policy.* 2002;14:27–39.
28. Kaufert JM, Putsch RW, Lavalley M. End-of-life decision making among Aboriginal Canadians: interpretation, mediation, and discord in the communication of “bad news.” *J Palliat Care.* 1999;15:31–38.
29. Newbold KB. Disability and use of support services within the Canadian aboriginal population. *Health Soc Care Community.* 1999;7:291–300.
30. Kelly L, Minty A. End-of-life issues for aboriginal patients: a literature review. *Can Fam Physician.* 2007;53:1459–1465.
31. McGrath P. Exploring Aboriginal people's experience of relocation for treatment during end-of-life care. *Int J Palliat Nurs.* 2006;12:102–108.
32. Brown EA, Bekker HL, Davison SN, et al. Supportive care: communication strategies to improve cultural competence in shared decision making. *Clin J Am Soc Nephrol.* 2016.
33. Kagawa-Singer M, Kassim-Lakha S. A strategy to reduce cross-cultural miscommunication and increase the likelihood of improving health outcomes. *Acad Med.* 2003;78:577–587.
34. Davison SN, Jhangri GS, Holley JL, et al. Nephrologists' reported preparedness for end-of-life decision-making. *Clin J Am Soc Nephrol.* 2006;1:1256–1262.
35. Gill TM, Gahbauer EA, Han L, et al. Trajectories of disability in the last year of life. *N Engl J Med.* 2010;362:1173–1180.
36. Chiu HH, Tangri N, Djurdjev O, et al. Perceptions of prognostic risks in chronic kidney disease: a national survey. *Can J Kidney Health Dis.* 2015;2:53.
37. Schell JO, Patel UD, Steinhilber KE, et al. Discussions of the kidney disease trajectory by elderly patients and nephrologists: a qualitative study. *Am J Kidney Dis.* 2012;59:495–503.
38. Couchoud C, Hemmelgarn B, Kotanko P, et al. Supportive care: time to change our prognostic tools and their use in CKD. *Clin J Am Soc Nephrol.* 2016.
39. Ritchie JP, Alderson H, Green D, et al. Functional status and mortality in chronic kidney disease: results from a prospective observational study. *Nephron Clin Pract.* 2014;128:22–28.
40. Painter P, Roshanravan B. The association of physical activity and physical function with clinical outcomes in adults with chronic kidney disease. *Curr Opin Nephrol Hypertens.* 2013;22:615–623.
41. Johansen KL, Chertow GM, Jin C, et al. Significance of frailty among dialysis patients. *J Am Soc Nephrol.* 2007;18:2960–2967.
42. Walker SR, Gill K, Macdonald K, et al. Association of frailty and physical function in patients with non-dialysis CKD: a systematic review. *BMC Nephrol.* 2013;14:228.
43. Lee SJ, Son H, Shin SK. Influence of frailty on health-related quality of life in pre-dialysis patients with chronic kidney disease in Korea: a cross-sectional study. *Health Qual Life Outcomes.* 2015;13:70.
44. McAdams-DeMarco MA, Law A, Salter ML, et al. Frailty as a novel predictor of mortality and hospitalization in individuals of all ages undergoing hemodialysis. *J Am Geriatr Soc.* 2013;61:896–901.
45. Kutner NG, Zhang R, Huang Y, et al. Gait speed and mortality, hospitalization, and functional status change among hemodialysis patients: a US renal data system special study. *Am J Kidney Dis.* 2015;66:297–304.
46. Usvyat LA, Kooman JP, van der Sande FM, et al. Dynamics of hospitalizations in hemodialysis patients: results from a large US provider. *Nephrol Dial Transplant.* 2014;29:442–448.
47. Johnstone S, Dombro L, Garza GSCK, et al. Declines in hemodialysis patients' physical and mental component scores before death. *J Am Soc Nephrol.* 2014;25.
48. Thijssen S, Wong MM, Usvyat LA, et al. Nutritional competence and resilience among hemodialysis patients in the setting of dialysis initiation and hospitalization. *Clin J Am Soc Nephrol.* 2015;10:1593–1601.
49. Kalantar-Zadeh K, Block G, McAllister CJ, et al. Appetite and inflammation, nutrition, anemia, and clinical outcome in hemodialysis patients. *Am J Clin Nutr.* 2004;80:299–307.

50. Kotanko P, Kooman J, van der Sande F, et al. Accelerated or out of control: the final months on dialysis. *J Ren Nutr*. 2014;24:357–363.
51. Kotanko P, Thijssen S, Usvyat L, et al. Temporal evolution of clinical parameters before death in dialysis patients: a new concept. *Blood Purif*. 2009;27:38–47.
52. Stacey D, Legare F, Col NF, et al. Decision aids for people facing health treatment or screening decisions. *Cochrane Database Syst Rev*. 2014;(1):CD001431.
53. Davison SN, Jhangri GS, Johnson JA. Cross-sectional validity of a modified Edmonton symptom assessment system in dialysis patients: a simple assessment of symptom burden. *Kidney Int*. 2006;69:1621–1625.
54. Davison SN, Jhangri GS, Johnson JA. Longitudinal validation of a modified Edmonton symptom assessment system (ESAS) in haemodialysis patients. *Nephrol Dial Transplant*. 2006;21:3189–3195.
55. Davison SN, Jhangri GS. Impact of pain and symptom burden on the health-related quality of life of hemodialysis patients. *J Pain Symptom Manage*. 2010;39:477–485.
56. Kimmel PL, Emont SL, Newmann JM, et al. ESRD patient quality of life: symptoms, spiritual beliefs, psychosocial factors, and ethnicity. *Am J Kidney Dis*. 2003;42:713–721.
57. Edmonton Zone Palliative Care Program and Northern Alberta Renal Program. 2016. <http://www.palliative.org/tools.html>. Accessed March 31, 2018.
58. Davison SN, Koncicki H, Brennan F. Pain in chronic kidney disease: a scoping review. *Semin Dial*. 2014;27:188–204.
59. Dworkin RH, Turk DC, Wyrwich KW, et al. Interpreting the clinical importance of treatment outcomes in chronic pain clinical trials: IMMPACT recommendations. *J Pain*. 2008;9:105–121.
60. Andrew Moore R, Eccleston C, Derry S, et al. “Evidence” in chronic pain—establishing best practice in the reporting of systematic reviews. *Pain*. 2010;150:386–389.
61. Joshi VD, Mooppil N, Lim JF. Validation of the kidney disease quality of life-short form: a cross-sectional study of a dialysis-targeted health measure in Singapore. *BMC Nephrol*. 2010;11:36.
62. Boyce MB, Browne JP, Greenhalgh J. The experiences of professionals with using information from patient-reported outcome measures to improve the quality of healthcare: a systematic review of qualitative research. *BMJ Qual Saf*. 2014;23:508–518.
63. Antunes B, Harding R, Higginson IJ. Implementing patient-reported outcome measures in palliative care clinical practice: a systematic review of facilitators and barriers. *Palliat Med*. 2014;28:158–175.
64. Laupacis A, Wong C, Churchill D. The use of generic and specific quality-of-life measures in hemodialysis patients treated with erythropoietin. The Candaian Erythropoietin Study Group. *Control Clin Trials*. 1991;12:168S–179S.
65. Gibbons E, Fitzpatrick R. *A Structured Review of Patient-Reported Outcome Measures for People with Chronic Kidney Disease*. University of Oxford; 2009.
66. Dowell D, Haegerich TM, Chou R. CDC guideline for prescribing opioids for chronic pain—United States, 2016. *JAMA*. 2016;315:1624–1645.
67. Kidney Supportive Care Research Group. *Conservative Kidney Management Care Pathway*; 2016. www.ckmcare.com. Accessed September 20, 2017.
68. Davison SN, Tupala B, Wasyluk BA, et al. *Recommendations for the care of patients receiving conservative kidney management: Focus on management of chronic kidney disease and symptoms*. CJASN; 2018 (in press).
69. Merskey H, Bogduk N. *Classification of Chronic Pain*. Seattle: International Association for the Study of Pain Press; 1994;210.
70. Davison SN. Facilitating advance care planning for patients with end-stage renal disease: the patient perspective. *Clin J Am Soc Nephrol*. 2006;1:1023–1028.
71. Davison SN, Torgunrud C. The creation of an advance care planning process for patients with ESRD. *Am J Kidney Dis*. 2007;49:27–36.
72. Fried TR, Bullock K, Iannone L, et al. Understanding advance care planning as a process of health behavior change. *J Am Geriatr Soc*. 2009;57:1547–1555.
73. Davison SN. Advance care planning in patients with end-stage renal disease. *Prog Palliat Care*. 2009;17:170–178.
74. Chambers EJ, Brown E, Germain M. *Supportive Care for the Renal Patient*. Oxford University Press; 2010.
75. Fried TR, Redding CA, Robbins ML, et al. Promoting advance care planning as health behavior change: development of scales to assess decisional balance, medical and religious beliefs, and processes of change. *Patient Educ Couns*. 2012;86:25–32.
76. Schell JO, Arnold RM. NephroTalk: communication tools to enhance patient-centered care. *Semin Dial*. 2012;25:611–616.
77. Schell JO, Green JA, Tulskey JA, et al. Communication skills training for dialysis decision-making and end-of-life care in nephrology. *Clin J Am Soc Nephrol*. 2013;8:675–680.
78. Murtagh FE, Marsh JE, Donohoe P, et al. Dialysis or not? A comparative survival study of patients over 75 years with chronic kidney disease stage 5. *Nephrol Dial Transplant*. 2007;22:1955–1962.
79. Smith C, Da Silva-Gane M, Chandna S, et al. Choosing not to dialyze: evaluation of planned non-dialytic management in a cohort of patients with end-stage renal failure. *Nephron Clin Pract*. 2003;95:c40–c46.
80. Foote C, Kotwal S, Gallagher M, et al. Survival outcomes of supportive care versus dialysis therapies for elderly patients with end-stage kidney disease: a systematic review and meta-analysis. *Nephrology (Carlton)*. 2016;21:241–253.
81. Murtagh FE, Burns A, Moranne O, et al. Supportive care: comprehensive conservative care in end-stage kidney disease. *Clin J Am Soc Nephrol*. 2016.
82. Tamura MK, Covinsky K, Chertow G, et al. Functional status of elderly adults before and after initiation of dialysis. *N Engl J Med*. 2009;361:1539–1547.
83. Jassal SV, Chiu E, Hladunewich M. Loss of independence in patients starting dialysis at 80 years of age or older. *N Engl J Med*. 2009;361:1612–1613.
84. O'Hare AM, Rodriguez RA, Hailpern SM, et al. Regional variation in health care intensity and treatment practices for end-stage renal disease in older adults. *JAMA*. 2010;304:180–186.
85. Ladin K, Lin N, Hahn E, et al. Engagement in decision-making and patient satisfaction: a qualitative study of older patients' perceptions of dialysis initiation and modality decisions. *Nephrol Dial Transplant*. 2016.
86. Song MK, Lin FC, Gilet CA, et al. Patient perspectives on informed decision-making surrounding dialysis initiation. *Nephrol Dial Transplant*. 2013;28:2815–2823.
87. O'Connor AM, Jacobsen MJ. *Workbook on Developing and Evaluating Patient Decision Aids*. Ottawa, ON: The Ottawa Hospital Research Institute; 2003.
88. Legare F, Ratte S, Stacey D, et al. Interventions for improving the adoption of shared decision making by healthcare professionals. *Cochrane Database Syst Rev*. 2010;CD006732.
89. Sepucha KR, Borkhoff CM, Lally J, et al. Establishing the effectiveness of patient decision aids: key constructs and measurement instruments. *BMC Med Inform Decis Mak*. 2013;13(suppl 2):S12.
90. Davis JL, Davison SN. Hard choices, better outcomes: a review of shared decision-making and patient decision aids around dialysis initiation and conservative kidney management. *Curr Opin Nephrol Hypertens*. 2017;26:205–213.
91. Murphy E, Germain MJ, Cairns H, et al. International variation in classification of dialysis withdrawal: a systematic review. *Nephrol Dial Transplant*. 2014;29:625–635.
92. Lambie M, Rayner HC, Bragg-Gresham JL, et al. Starting and withdrawing haemodialysis—associations between nephrologists' opinions, patient characteristics and practice patterns (data from the dialysis outcomes and practice patterns study). *Nephrol Dial Transplant*. 2006;21:2814–2820.
93. Tangri N, Stevens LA, Griffith J, et al. A predictive model for progression of chronic kidney disease to kidney failure. *JAMA*. 2011;305:1553–1559.
94. Tangri N, Grams ME, Levey AS, et al. Multinational assessment of accuracy of equations for predicting risk of kidney failure: a meta-analysis. *JAMA*. 2016;315:164–174.
95. Thamer M, Kaufman JS, Zhang Y, et al. Predicting early death among elderly dialysis patients: development and validation of a risk score to assist shared decision making for dialysis initiation. *Am J Kidney Dis*. 2015;66:1024–1032.
96. Couchoud CG, Beuscart JB, Aldigier JC, et al. Development of a risk stratification algorithm to improve patient-centered care and decision making for incident elderly patients with end-stage renal disease. *Kidney Int*. 2015;88:1178–1186.
97. Mauri JM, Vela E, Cleries M. Development of a predictive model for early death in diabetic patients entering hemodialysis: a population-based study. *Acta Diabetol*. 2008;45:203–209.

98. Couchoud C, Labeeuw M, Moranne O, et al. A clinical score to predict 6-month prognosis in elderly patients starting dialysis for end-stage renal disease. *Nephrol Dial Transplant*. 2009;24:1553–1561.
99. Cohen LM, Ruthazer R, Moss AH, et al. Predicting six-month mortality for patients who are on maintenance hemodialysis. *Clin J Am Soc Nephrol*. 2010;5:72–79.
100. Murphy EL, Murtagh FE, Carey I, et al. Understanding symptoms in patients with advanced chronic kidney disease managed without dialysis: use of a short patient-completed assessment tool. *Nephron Clin Pract*. 2008;111:c74–c80.
101. Cleeland CS, Ryan KM. Pain assessment: global use of the Brief Pain Inventory. *Ann Acad Med Singapore*. 1994;23:129–138.
102. Dworkin RH, Turk DC, Farrar JT, et al. Core outcome measures for chronic pain clinical trials: IMMPACT recommendations. *Pain*. 2005;113:9–19.
103. Kutner JS, Blatchford PJ, Taylor DH Jr, et al. Safety and benefit of discontinuing statin therapy in the setting of advanced, life-limiting illness: a randomized clinical trial. *JAMA Intern Med*. 2015;175:691–700.
104. Benetos A, Rossignol P, Cherubini A, et al. Polypharmacy in the aging patient: management of hypertension in octogenarians. *JAMA*. 2015;314:170–180.
105. Muller M, Smulders YM, de Leeuw PW, et al. Treatment of hypertension in the oldest old: a critical role for frailty? *Hypertension*. 2014;63:433–441.
106. Musini VM, Tejani AM, Bassett K, et al. Pharmacotherapy for hypertension in the elderly. *Cochrane Database Syst Rev*. 2009;CD000028.
107. Kopple JD, Kalantar-Zadeh K, Mehrotra R. Risks of chronic metabolic acidosis in patients with chronic kidney disease. *Kidney Int Suppl*. 2005;S21–S27.

BOARD REVIEW QUESTIONS

1. A 62-year-old woman on hemodialysis presents with a 6-month history of a constant dull 6/10 pain in her shoulders and knees. What medication would you use to control her pain initially while investigating the cause of her pain?
- Gabapentin 100 mg nightly
 - Acetaminophen 500–1000 mg three times daily
 - Nonsteroidal antiinflammatory at a low dose two times daily
 - Hydromorphone 0.5–1.0 mg every 4–6 hours as needed
 - No analgesic should be prescribed until the cause of pain is identified and reversible factors addressed

Answer: b

Rationale: This woman has nociceptive pain and can be treated with traditional analgesics, rather than an adjuvant such as gabapentin. The first line of therapy is the nonopioid acetaminophen, even though her initial pain is moderate in severity. It is not necessary to wait for a diagnosis to initiate therapy. One should try to initiate analgesia before the pain becomes intolerable and established, as pain is more difficult to treat once the pain cycle becomes established.

2. A 75-year-old man with an estimated glomerular filtration rate of 9 mL/min/1.73 m² being managed with conservative kidney management has severe chronic pain from inoperable small vessel peripheral vascular disease that he rates as high as 7/10. What medications would you use to initiate treatment for his pain?
- Gabapentin 100 mg nightly
 - Acetaminophen 1000 mg three times daily
 - Gabapentin 100 mg nightly + acetaminophen 1000 mg three times daily
 - Hydromorphone 0.5–1.0 mg every 4–6 hours as needed
 - Gabapentin 100 mg nightly + hydromorphone 0.5–1.0 mg every 4–6 hours as needed

Answer: c

Rationale: Peripheral vascular disease almost always has a neuropathic component, which generally responds poorly

to opioids or requires doses for a response that are associated with unacceptable toxicity. Adjuvant analgesics such as gabapentin should be tried first. Given that the initial pain is severe, it is reasonable to add a nonopioid analgesic such as acetaminophen concurrently while the gabapentin is being titrated to maximal effectiveness. Only after the adjuvant analgesic has been titrated to maximal effect should the addition of opioids be considered if the pain remains inadequately controlled. Initiating opioids and gabapentin concurrently in an elderly patient with advanced chronic kidney disease and multimorbidity would put him at high risk from adverse effects including falls.

3. For whom might conservative kidney management (CKM) be a viable treatment option?
- An 84-year-old who is functionally independent and has a long-standing hypertension
 - A 67-year-old with multiple comorbidities including poor cognitive function
 - A 45-year-old diabetic with severe vascular disease who is not a transplant candidate and who does not wish to proceed with dialysis
 - None of the above
 - b and c

Answer: c

Rationale: Age alone is not an ideal criterion for the consideration of CKM. Elderly patients who remain robust are likely to have a survival advantage, even if it is small, with dialysis. Regardless of age, those who have multiple comorbidities, especially cognitive dysfunction, are not likely to benefit substantially from dialysis and CKM should be considered. Even in younger patients who through informed consent view the burdens of dialysis to outweigh the benefits, CKM is a reasonable option to consider.